BOUNDS FOR THE LENGTH OF FINITE PIERCE EXPANSIONS

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ABSTRACT. For integers $a > b > 0$, let $P(a,b)$ count the positive terms of the iteration $x \mapsto a \bmod x$, starting at b. We prove

$$P(a,b) \ll_\varepsilon a^{\alpha_*+\varepsilon}, \qquad \alpha_* = (1310 - \sqrt{655})/4365 = 0.2942513246\ldots.$$

Each tail of the orbit carries a finite positive measure whose Hankel determinants are positive integers. We group short segments of the orbit by small values of these determinants. Within a group, the division equations $a = q_i b_i + b_{i+1}$, kept with the fixed dividend a, confine the quotient increments to a union of boundedly many affine curves. Adjoining two further integer determinants as coordinates raises the degree of these curves while keeping the coordinates small. A curve-counting theorem whose constants do not depend on the coefficients of the curve then bounds the number of segments in each group. Raising the added coordinates to suitable powers and balancing their sizes gives the exponent α_*. We also prove several elementary intermediate bounds, a bound for orbits with few rises in their quotient increments, and positivity and integrality of the Hankel determinants. The main result and its curve-counting input are formalized in Lean 4 with Mathlib without proof holes or additional axioms. A separate refinement $P(a,b) \ll a^{3/10}\sqrt{\log\log 3a}$ is formalized with two published analytic estimates as explicit hypotheses. Shallit's conjectured $O((\log a)^2)$ bound remains open.

1. INTRODUCTION

Let $a > b > 0$ be integers. Put $b_0 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence is strictly decreasing until it reaches 0; if $b_n = 0$ we write $P(a,b) = n$. For example $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. The number $P(a,b)$ is the length of the Pierce expansion of b/a [12, 13], since

$$\frac{b}{a} = \frac{1}{q_0} - \frac{1}{q_0 q_1} + \frac{1}{q_0 q_1 q_2} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [11] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [13] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [6] combined these with a divisor argument to get $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [5] went past the exponent $1/3$, proving $P(a,b) \ll_\varepsilon a^{1/3-2/177+\varepsilon}$ with exponential sum estimates (Bourgain's exponent pair), and showed that $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Baraskar and Vukusic [1] study the related sets of iterated remainders and bound their cardinalities at each fixed iteration depth. Shallit's talk pages [16] report a bound $n^{3/10}$, described as "LLM, unpublished so far". We have not seen that argument. The preliminary $3/10$ theorem below has the same exponent up to ε, and we make no claim about how the two arguments are related.

Our strongest bound comes from the joint-invariant curve embedding of Section 15.

Theorem 1.1. *For every $\varepsilon > 0$ there is $C_\varepsilon > 0$, independent of a, b, such that*

$$P(a,b) \le C_\varepsilon a^{(1310-\sqrt{655})/4365+\varepsilon} \qquad (0 < b < a).$$

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Here $(1310 - \sqrt{655})/4365 = 0.2942513\ldots$. The proof produces, for each pair of positive integers (m, n), an explicit rational exponent (Corollary 15.16); the pairs $(1, 1)$ and $(50, 51)$ give $128/435$ and $2201/7480$, which are slightly larger than α_*. The formalisation states these two instances separately, alongside Theorem 1.1.

The joint-invariant argument. We count orbit terms in a remainder window $[A, 2A)$ and then sum over dyadic values of A; let T be the number of terms in the window. For most indices i in the window, the next few quotient increments $k_j = q_{i+j+1} - q_{i+j}$ are at most $H \asymp a/(AT)$. Each tail of the orbit carries a positive measure (Section 10) whose Hankel determinants $I_r(i)$ are positive integers; I_1, I_2 and I_3 are the remainder, the drop, and the curvature D of three consecutive steps (Section 3). The ten 3×3 minors of an explicit 3×5 integer matrix of drops and increments have size at most $X \asymp A^4H^6/a^3$ (Section 13). Their vector is the *pattern* of i, and there are $a^{o(1)}X^5$ possible patterns.

Fix a pattern. The increments are then determined by the three *nodes* $s_j = q_{i+j} - q_i$, $j = 1, 2, 3$, which are integers in $[0, 3H]$. The nodes satisfy a quadratic equation coming from linear relations among the minors. Keeping the fixed dividend a, they also satisfy the degree-twelve equation $N_0 \operatorname{Res}(P_4, xR_4) = a\Psi^4$, where $N_0 = I_4(i)$, P_4 and R_4 are the degree-four orthogonal polynomials of two consecutive tails, and Ψ is the squared norm of P_4 (Lemma 15.1). So the nodes lie on at most 24 curves, each of degree at most 24. Counting integer points on these curves in the box $[0, 3H]^3$ directly saves at most a factor $H^{1/2}$ per curve, because a component may have degree two.

To do better, we adjoin further integer coordinates: three 2×2 minors $W = (W_0, W_1, W_2)$ of drops and increments, one of which is the curvature, of size at most $S \asymp A^3H^3/a^2$; and the fifth Hankel determinant $J = I_5(i)$, of size at most $Y \asymp A^5H^{10}/a^4$. On a component C of degree d these are rational functions of the nodes. At a point of C at infinity of multiplicity e, the W_j vanish to order $3e/2$ and J has a pole of order $5e/2$. A linear combination of the W_j also has one zero at an affine point, which forces one extra affine pole. Consequently the curve obtained by adjoining W_0^m, W_1^m, W_2^m, J^n to the nodes has degree at least $10n + 7m$, while its coordinates are at most $\max(3H, S^m, Y^n)$. The bound of Castryck, Cluckers, Dittmann and Nguyen [3] counts the integer points of a curve of degree δ in a box of side B by $B^{1/\delta+o(1)}$, uniformly in the coefficients of the curve. Applied to the enlarged curves, it bounds the number of starts per pattern. Optimising over the ratio n/m gives α_* (Section 15).

The number of patterns, $a^{o(1)}X^5$, sets a floor for this method. Even as $m, n \to \infty$, the pattern count alone gives $T \ll a^{5/17+o(1)}$, and α_* exceeds $5/17 = 0.2941176\ldots$ by only $1.3 \cdot 10^{-4}$ (Section 17).

Reading guide. Sections 3–7 contain the elementary window bounds, the six-step identity and the proof of Theorem 1.2. The proof of Theorem 1.1 then uses:

- the moment model and Theorem 10.2 (Section 10);
- the pattern matrix and Lemmas 11.1, 11.4 and 11.5, and the good starts of Proposition 11.8 (Section 11);
- only Lemma 12.1 from Section 12;
- the minor vectors of Section 13 (Lemma 13.1 and Propositions 13.2 and 13.3);
- and Section 15 itself.

The short-interval divisor refinement (Section 8), the rise bound (Section 9), the slice and norm-equation counts of Section 12 and the lattice refinement of Section 14 are not needed for Theorem 1.1. Section 16 describes the formal verification and Section 17 the limits of the method.

Elementary window bounds. The simplest of our bounds is the following.

Theorem 1.2. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{3/10+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

The proof is elementary. It uses only the division equations $a = q_i b_i + b_{i+1}$ and the divisor bound $\tau(n) \ll_\delta n^\delta$. As usual, the remainders are split into dyadic windows $[A, 2A)$, and we bound the number T of remainders in one window. There are two bounds:

$$T(T-1) \le 4A\,\Delta, \tag{2}$$

$$T^6 A^2 \ll a^3 \Delta, \tag{3}$$

where Δ is the largest value of the divisor function on $[1, 32a^7]$. Bound (2) is the Erdős–Shallit argument. Bound (3) is new. Multiplying (3) by the square of (2) eliminates A and gives $T^{10} \ll a^3\Delta^3$, which is the exponent $3/10$.

The new bound comes from three consecutive steps $x = b_i$, $y = b_{i+1}$, $z = b_{i+2}$, $w = b_{i+3}$. Let d, e, f be the drops, let u, v be the increments of the quotients, and put $D = dv - eu$. Then

$$D\,z = df + ude + (u-1)e^2. \tag{4}$$

So D is a *positive* integer, even though dv and eu largely cancel, and it is small when the drops are small. Also $(q, x) = (\lfloor a/b_i \rfloor, b_i)$ lies on the curve $xM(q) = aN(q) + D$, where M and N are explicit polynomials depending on (u, v) (Lemma 3.3). A resultant-type identity shows that each such curve has $a^{o(1)}$ admissible points (Lemma 4.1). The number of triples (u, v, D) is $\ll (B/T)^2 \cdot a/T^3$, where $B = a/A$, which gives (3).

A second positive integer. Six consecutive terms carry three overlapping curvatures D, E, F and four quotient increments u, v, w, z. In Section 7 we show that the integer

$$L = \frac{S(v,w,z)\,D - S(u,v,w)\,E}{v}, \qquad S(u,v,w) = u^2v + (u-1)v^2 + uw,$$

satisfies the exact identity $qwL = J(u,v,w,z)\,E + S(u,v,w)\,F$, where J is a polynomial with positive values (Lemma 7.1). Hence $L \ge 1$, and $q \asymp a/A$ is bounded by a polynomial in the small parameters u, v, w, z, E, F. A single window of six terms with small total drop then gives

$$T^6 A^2 \ll a^3, \tag{5}$$

with no divisor function at all (Proposition 7.3).

With (5) in hand, the only remaining loss is in the drop bound (2). That loss comes from the number of divisors of $a+h$ in $[A, 2A)$ as h ranges over a short interval. Section 8 shows that any bound $\ll y\,G(x)$ for such counts over intervals $(x, x+y]$ with $y \ge x^{1/4}$, uniform in A and with $G(x) \ll x^{1/40}$, gives $P(a,b) \ll a^{3/10} G(a)^{1/5}$ (Theorem 8.4). The counts are majorised by the Erdős–Hooley function. Combining the short-sum theorem of Nair and Tenenbaum [9] with the recent mean-value bound of de la Bretèche and Tenenbaum [2] gives $G(x) = (\log\log x)^{5/2}$, and hence the following.

Theorem 1.3. *Assume the theorem of Nair–Tenenbaum stated as* (9) *and the theorem of de la Bretèche–Tenenbaum stated as* (10) *below. Then*

$$P(a,b) \ll a^{3/10}\sqrt{\log\log 3a} \qquad \text{uniformly for integers } 0 < b < a.$$

Both hypotheses are published theorems. They are hypotheses only in the sense of the formalisation: we have not formalised their proofs. Any other short-interval divisor bound can be used in their place. For example, Shiu's theorem [15] gives $\sum_{x<n\le x+y} \tau(n) \ll y \log x$ for $y \ge x^{1/4}$, and hence $P(a,b) \ll a^{3/10}(\log a)^{1/5}$; we have not formalised that either.

Structured trajectories and the Hankel hierarchy. Write $k_i = q_{i+1} - q_i$ for the quotient increments, and put

$$R = 1 + \#\{i : 0 \le i \le P(a,b) - 3,\ k_i < k_{i+1}\}.$$

Thus $R - 1$ counts the rises between consecutive increments.

Theorem 1.4. *$P(a,b) \ll a^{1/4}\sqrt{R}$ uniformly for $0 < b < a$. In particular, if the quotient increments never rise, then $P(a,b) \ll a^{1/4}$.*

The proof (Section 9) combines three facts. First, $D \ge 1$ says that d_i/k_i is strictly decreasing, so the drops strictly decrease along every stretch without a rise. Second, the six-step integer L gives an exact identity for six quotients in arithmetic progression, which forces their common difference to be large (Lemma 9.1). Third, a sequence of positive integers in which no value occurs too often has a large sum (Lemma 9.3). Conversely, Theorem 1.4 shows that an orbit of length about $a^{3/10}$ must contain $\gg a^{1/10}$ rises.

Finally, both D and L fit into a single structure (Section 10). The orbit carries an exact finite model by positive measures ν_i with $\int d\nu_i = b_i/a$. The Hankel determinants $I_r(i) = a \det\bigl(\int x^{j+l}\, d\nu_i\bigr)_{j,l<r}$ are positive integers whenever the remaining orbit has at least r positive terms. The first four agree with b_i, $b_i - b_{i+1}$, D and L in the nonterminal ranges specified in Theorem 10.2.

Below 3/10. The following four bounds lie between Theorems 1.2 and 1.1. They introduce, one at a time, the pattern-counting ingredients that Theorem 1.1 builds on, together with a lattice refinement that it does not need.

Theorem 1.5. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{23/77+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $23/77 = 0.298701\ldots$, a saving of $1/770$ over $3/10$. The proof (Section 11) uses the fourth Hankel invariant $I_4 = L$ as an integer, not only as a positive number. At the critical configuration L has bounded size. The labels $L(i), L(i+1), L(i+2)$ and four cofactors of an explicit 3×4 integer matrix form a *pattern* of seven integers. Two consecutive states give two linear equations and one quadratic equation in five consecutive quotient increments, and the quadratic has a positive leading coefficient. An exact identity $I_4(i)\,\pi_i(-q_i) = a \det(\mu_i(x^{j+l}))_{j,l<4}$, with π_i the degree-four orthogonal polynomial of the moment model, makes q_i a root of a quartic that is fixed by the pattern and the increments. So each pattern accounts for $O(H^2)$ states, where H bounds the increments. The number of patterns is controlled by Hadamard's inequality applied to the moment model.

Counting the solutions of the quadratic equation more carefully gives a further improvement.

Theorem 1.6. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{11/37+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $11/37 = 0.297297\ldots$, a saving of $1/370$ over $3/10$. The extra input (Section 12) is that the quadratic part is nondegenerate with an indefinite binary restriction, by exact sign identities for the cofactors. So each slice is a binary norm equation, which has $a^{o(1)}$ solutions of size $a^{O(1)}$, uniformly in its coefficients, when its right-hand side is nonzero. There are at most two exceptional slices, which are counted separately.

Theorem 1.7. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{8/27+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $8/27 = 0.296296\ldots$. The seven pattern integers are signed coordinates of the vector of 3×3 minors of a 3×5 integer matrix (Section 13). All ten minors are bounded by X, and such minor vectors number $X^{5+o(1)}$ rather than X^7. From Section 13 on, the *pattern* of a state means this vector of ten minors, which determines the seven integers above.

Theorem 1.8. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{45/152+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

Here $45/152 = 0.296052\ldots$. The pivot gaps of a state lie in a lattice $\Lambda_p \subseteq \mathbb{Z}^3$ of index at most X determined by its minor vector p (Section 14). If the shortest vector of Λ_p has length λ, a reduced basis cuts the ternary quadratic into $O(H/\lambda+1)$ slices, and only $X^{4+o(1)}t^3$ minor vectors have $\lambda \le t$. This gives $T \ll a^{o(1)}(HX^{14/3} + X^5)$ in place of $T \ll a^{o(1)}HX^5$.

The joint-invariant argument returns to the ordinary minor-pattern count of Section 13 and improves the count within each pattern. Its additional input is the affine-curve theorem used in Section 15.

Formal verification. Every step of the proofs of Theorems 1.1, 1.2, 1.5, 1.6, 1.7 and 1.8 has been formalised in Lean 4 [7] using Mathlib [8]. So have Proposition 7.3, Theorem 8.4, the deduction of Theorem 1.3 from (9) and (10), Theorem 1.4 and Theorem 10.2. Section 16 describes the formalisation. The constants in this paper are the constants in the Lean proof. The accompanying repository contains the proof source, independent submission statements and build instructions.

What is not proved. These results are far from (1), and the method cannot reach it; see Section 17.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_0 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a,b)$, so that $a > b_0 > b_1 > \cdots > b_{P-1} > 0 = b_P$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \le b_{i+1} < b_i \qquad (0 \le i < P).$$

An index i of the orbit is also called a *state*; quantities "at state i" are computed from $b_i, b_{i+1}, \ldots$ and $q_i, q_{i+1}, \ldots$, and are defined whenever the orbit terms they use exist. Let $\tau(n)$ be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1 \le n \le 32a^7} \tau(n).$$

We write $F \ll_\eta G$ when $F \le C_\eta G$, with the constant uniform in the dividend and orbit. A factor denoted $a^{o(1)}$ is bounded by $C_\eta a^\eta$ for every fixed $\eta > 0$; such factors include the divisor losses below. We write $F \asymp G$ when both $F \ll G$ and $G \ll F$ hold. Many statements below carry explicit numerical constants; these are the constants of the Lean proof. Where a constant is large we state the bound with $\ll$ and record the constant in the proof.

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \ge q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i+1)b_{i+1} \le q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1} \rfloor \ge q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \ge 1$ there is C_k with $\tau(n)^k \le C_k\, n$ for all $n \ge 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k/p^\alpha$. Each factor with $p \ge 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \le 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \geq 1$ be an integer. A *window run* of length T is a sequence $x_0, x_1, \ldots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \leq a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \geq 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \geq 1$, and clearly $d, e, f \geq 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{6}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \geq 1$.

Proof. Subtracting the first two equations of (6) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \geq 1$, and $z > 0$, so $D > 0$. □

Lemma 3.3 (The fiber equation)**.** *Put* $M(q) = uq^2 + (u^2+u+v)q + v$ *and* $N(q) = uq + u^2 + v$. *Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (6), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If* $x \leq a$ *then*

$$u\,yx \leq a(d+e), \qquad v\,zy \leq a(e+f), \qquad D\,z \leq (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \leq ad + ae$, since $qx \leq a$ and $x \leq a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \leq a$ and $y \leq a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \leq df + u\,e(d+e) \leq (1+u)(d+e+f)^2$. □

4. Few blocks share a triple (u, v, D)

Lemma 4.1 (Fiber count)**.** *Let $a \geq 1$, $1 \leq u, v \leq a$ and $1 \leq D \leq 2a$. Let $M(q) = uq^2 + pq + v$ with $p = u^2 + u + v$, and let $\Lambda(q) = cq + \ell$ with $c = au$ and $\ell = a(u^2 + v) + D$. Then the set*

$$Q = \{q \in \mathbb{Z}_{\geq 0} : \ xM(q) = \Lambda(q) \text{ for some integer } x \geq 1\}$$

has at most $\Delta = \max_{1 \leq n \leq 32a^7} \tau(n)$ elements.

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)\Lambda(q) = R \tag{7}$$

identically in q. If $q \in Q$ with witness x, substituting $\Lambda(q) = xM(q)$ into (7) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{8}$$

We bound the sizes first. From $u, v \leq a$ and $D \leq 2a$ we get $\ell \leq 4a^3$, $p \leq 3a^2$ and $c \leq a^2$, so

$$|R| \leq u\ell^2 + cp\ell + c^2 v \leq 16a^7 + 12a^7 + a^5 \leq 32a^7, \qquad c^2 \leq a^4 \leq 32a^7.$$

Case $R \neq 0$. By (8), $M(q)$ is a positive divisor of $|R|$. Since $u, p \geq 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \geq 0$, hence injective. Therefore $|Q| \leq \tau(|R|) \leq \Delta$.

Case $R = 0$. Since $M(q) > 0$, (8) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \geq 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \leq \tau(c^2) \leq \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and Λ might share a factor.

5. Counting visits to a window

In this section $x_0, \ldots, x_{T-1}$ is a window run in $[A, 2A)$, and $q_j = \lfloor a/x_j \rfloor$ are its quotients. We continue to write $\Delta = \max_{1 \leq n \leq 32a^7} \tau(n)$.

Proposition 5.1 (Drop bound; after Erdős–Shallit)**.** *If $T \geq 2$ then $T(T-1) \leq 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \geq 1$ for $0 \leq j \leq T-2$. By telescoping, $\sum_j h_j = x_0 - x_{T-1} \leq (2A-1) - A = A - 1$. Call j *good* if $h_j(T-1) \leq 2A$ and *bad* otherwise. Each bad j has $h_j(T-1) \geq 2A+1$, so

$$|\,\mathrm{Bad}\,|\,(2A+1) \leq (T-1)\textstyle\sum_j h_j \leq (T-1)(A-1).$$

If $2|\,\mathrm{Bad}\,| \geq T-1$ this would give $(T-1)(2A+1) \leq 2(T-1)(A-1)$, which is false. Hence $2|\,\mathrm{Bad}\,| < T-1$, and since $|\,\mathrm{Good}\,| + |\,\mathrm{Bad}\,| = T-1$ we get $2|\,\mathrm{Good}\,| \geq T$.

For good j we have $1 \leq h_j \leq H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j + 1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \leq 2a \leq 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a + h) \leq \Delta$ good indices have $h_j = h$. Hence $|\,\mathrm{Good}\,| \leq H\Delta$, and

$$T(T-1) \leq 2|\,\mathrm{Good}\,|(T-1) \leq 2\Delta\, H(T-1) \leq 4A\Delta.$$ □

Proposition 5.2 (Curvature bound)**.** *If $T \geq 12$ then $T^6 A^2 \leq 995328\, a^3 \Delta$.*

Proof. For $0 \leq j \leq T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \leq 3(2A-1) - 3A = 3A - 3.$$

Call j *good* if $s_j T \le 12A$ and *bad* otherwise. Then $|\operatorname{Bad}|(12A+1) \le T(3A-3)$. If $4|\operatorname{Bad}| \ge T$ this would give $T(12A+1) \le 4T(3A-3)$, which is false. So $4|\operatorname{Bad}| \le T-1$. Since $|\operatorname{Good}| = T - 3 - |\operatorname{Bad}|$ and $T \ge 12$,

$$2|\operatorname{Good}| \ge 2T - 6 - \tfrac{T-1}{2} \ge T.$$

Fix a good j. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d+e+f$. Lemma 3.4 gives $uA^2 \le uyx \le a(d+e) \le as$. Multiplying by T and using $sT \le 12A$ gives $uA^2T \le 12aA$, that is,

$$u\,AT \le 12a, \qquad \text{and likewise} \qquad v\,AT \le 12a.$$

Also $DA \le Dz \le (u+1)s^2 \le 2us^2$. Multiplying by T^2 gives $DAT^2 \le 2u(sT)^2 \le 288uA^2$, so $DT^2 \le 288uA$, and then

$$D\,T^3 \le 288\,u\,AT \le 3456\,a.$$

Thus the triple (u, v, D) of a good block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \le a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \le 2a.$$

The inequalities $U \le a$ and $D^* \le 2a$ hold because $AT \ge 12$ and $T^3 \ge 1728$.

Now fix such a triple and consider the good blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on these blocks: q_j determines $x_j = \Lambda(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ good blocks share a triple. So $|\operatorname{Good}| \le U^2 D^* \Delta$, and

$$T^6A^2 = T\,(AT)^2\,T^3 \le 2\Delta\,(U\,AT)^2\,(D^*T^3) \le 2\Delta(12a)^2(3456a) = 995328\,a^3\Delta. \qquad \square$$

Corollary 5.3 (Window bound)**.** *If $T \ge 12$ then $T^{10} \le 63700992\,a^3\Delta^3$.*

Proof. Using $T \le 2(T-1)$ and the two propositions,

$$T^{10}A^2 \le 4\,(T^6A^2)\,\big(T(T-1)\big)^2 \le 4 \cdot 995328\,a^3\Delta \cdot (4A\Delta)^2 = 63700992\,a^3\Delta^3A^2.$$

Divide by A^2. $\square$

Remark 5.4. In terms of $B = a/A$, Propositions 5.1 and 5.2 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 5.3 takes the product of the two bounds, so no case split is needed.

6. Proof of Theorem 1.2

Let $0 < b < a$. For $0 \le k \le \log_2 a$ let T_k be the number of indices $i < P$ with $2^k \le b_i < 2^{k+1}$. Every b_i with $i < P$ lies in $[1, a)$, so $P = \sum_{0 \le k \le \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

Let $\varepsilon > 0$. Choose an integer $m > 7/(5\varepsilon)$, and let C_{3m} be the constant of Lemma 2.2 with $k = 3m$. The maximum Δ is attained at some $n_0 \le 32a^7$, so $\Delta^{3m} \le C_{3m} \cdot 32a^7$. If $T_k \ge 12$, Corollary 5.3 gives

$$T_k^{10m} \le 63700992^m\,a^{3m}\,\Delta^{3m} \le K\,a^{3m+7}, \qquad K = 63700992^m \cdot C_{3m} \cdot 32,$$

and hence

$$T_k \le K^{1/(10m)}\,a^{3/10+7/(10m)} \le K^{1/(10m)}\,a^{3/10+\varepsilon/2}.$$

If $T_k \le 11$ this bound holds with 11 in place of $K^{1/(10m)}$. Finally $\lfloor \log_2 a \rfloor + 1 \le C'_\varepsilon a^{\varepsilon/2}$. Summing over k,

$$P(a,b) \le (\lfloor \log_2 a \rfloor + 1)\big(11 + K^{1/(10m)}\big)a^{3/10+\varepsilon/2} \le C_\varepsilon\,a^{3/10+\varepsilon}. \qquad \square$$

7. Six consecutive steps

Fix six consecutive terms $x_0 > x_1 > \cdots > x_5 \geq 0$ of the sequence (b_i) with $x_4 > 0$, and write their quotients as

$$\lfloor a/x_0 \rfloor = q, \quad \lfloor a/x_1 \rfloor = q+u, \quad \lfloor a/x_2 \rfloor = q+u+v,$$
$$\lfloor a/x_3 \rfloor = q+u+v+w, \quad \lfloor a/x_4 \rfloor = q+u+v+w+z.$$

(Unlike in Section 3, here w and z denote quotient increments.) By Lemma 2.1, $u, v, w, z \geq 1$. Let $f_k = x_k - x_{k+1} \geq 1$ be the drops, and let

$$D = f_0 v - f_1 u, \qquad E = f_1 w - f_2 v, \qquad F = f_2 z - f_3 w$$

be the curvatures of the blocks starting at x_0, x_1 and x_2. Write $\mathcal{E}_k$ for the division equation $x_k q_k + x_{k+1} - a = 0$, where $q_k = \lfloor a/x_k \rfloor$. Put

$$S(u,v,w) = u^2 v + (u-1)v^2 + uw,$$
$$J(u,v,w,z) = \big(u^2 + (u-1)v\big)z + w\big(u^3 + uv(2u-1) + (u-1)v^2\big) + w^2\big((u-1)(u+v)+1\big),$$
$$L = (vw + w^2 + z)\, D - \big(u^2 + (u-1)v\big)E - w\,(wf_0 - uf_2).$$

Lemma 7.1 (The six-step identity)**.** (1) $S(v,w,z)\, D - S(u,v,w)\, E = v\, L$.
(2) $q\, w\, L = J(u,v,w,z)\, E + S(u,v,w)\, F$.

Proof. Part (1) is a polynomial identity: since $S(v,w,z) \equiv -w^2$ and $S(u,v,w) \equiv uw \pmod{v}$, and $wD + uE = v(wf_0 - uf_2)$, the difference is divisible by v, and expanding gives L. Part (2) holds modulo the division equations. Explicitly, $qwL - JE - S(u,v,w)F$ equals

$$w(v^2 w + vw^2 + vz - w^2)\,\mathcal{E}_0 - w(u^2 w + 2uvw + uw^2 + uz + v^2 w + vw^2 - vw + vz - w^2)\,\mathcal{E}_1$$
$$+ w(u^2 v + u^2 w + uv^2 + 2uvw + uw^2 + uw + uz - v^2 - vw)\,\mathcal{E}_2 - w(u^2 v + uv^2 + uw - v^2)\,\mathcal{E}_3.$$

This is an identity of polynomials in a, q, the x_k and the increments, checked by expansion. □

Corollary 7.2. *Suppose $q \geq 1$ and $u, v, w, z \leq m$. Then $E, F \geq 1$, $L \geq 1$, and*

$$q \leq 9m^3 E + 3m^3 F.$$

Proof. Lemma 3.2, applied to the blocks starting at x_1 and x_2, gives $E, F \geq 1$. Clearly $J > 0$ and $S \geq 0$, so by Lemma 7.1(2) $qwL > 0$. Since L is an integer, $L \geq 1$ and $qw \leq JE + SF$. Bounding each monomial gives $J \leq 9m^3 w$ and $S(u,v,w) \leq 3m^3 w$. Dividing by w gives the claim. □

So the quotient q, which is about a/A, is controlled by the small parameters of a single block of six terms. No counting over many blocks is needed.

Proposition 7.3 (Six-step window bound)**.** *For a window run of length $T \geq 10$,*

$$T^6 A^2 \leq 96 \cdot 10^6\, a^3.$$

Proof. For $0 \leq j \leq T-6$ let $s_j = x_j - x_{j+5}$. By telescoping, $\sum_{j \leq T-6} s_j \leq 5(2A-1) - 5A = 5A-5$. So some j has $s_j(T-5) \leq 5A-5$, and since $T \leq 2(T-5)$, we get $s := s_j$ with $sT \leq 10A$. Apply Corollary 7.2 to $x_j, \ldots, x_{j+5}$, all of which lie in $[A, 2A)$, with $m = \max(u,v,w,z)$.

By Lemma 3.4 at x_j, x_{j+1} and x_{j+2}, each increment satisfies (increment) $\cdot\, A^2 \leq as$. Multiplying by T gives $mAT \leq 10a$. By Lemma 3.4 again, $EA \leq Ex_{j+3} \leq (v+1)s^2 \leq 2ms^2$, and likewise $FA \leq 2ms^2$. Hence $qA \leq 24m^4 s^2$. Also $a = qx_j + x_{j+1} < (q+1)x_j < 2A(q+1) \leq 4Aq$. Therefore

$$qA \cdot A^4 T^6 \leq 24\,(mAT)^4 (sT)^2 \leq 24 \cdot 10^6\, a^4 A^2,$$

so $qA^3 T^6 \leq 24 \cdot 10^6 a^4$, and $aA^2 T^6 \leq 4qA^3 T^6 \leq 96 \cdot 10^6\, a^4$. □

Remark 7.4. In terms of $B = a/A$, Proposition 7.3 reads $T \ll a^{1/6}B^{1/3}$. This is Proposition 5.2 without the factor $\Delta^{1/6}$. Combined with Proposition 5.1 as in Corollary 5.3, it gives $T^{10} \ll a^3\Delta^2$ and another proof of Theorem 1.2. The Lean proof of Theorem 1.2 uses Corollary 5.3.

8. Divisors over short intervals

For $A \geq 1$ let $\tau_A(n) = \#\{d \mid n : \ A \leq d < 2A\}$.

Proposition 8.1 (Drop count)**.** *For a window run of length $T \geq 2$ and any $Y \geq \lfloor 2A/(T-1) \rfloor$,*

$$T \leq 2 \sum_{a < n \leq a+Y} \tau_A(n).$$

Proof. This is the proof of Proposition 5.1, stopped before the divisor bound. The good indices j have $1 \leq h_j \leq \lfloor 2A/(T-1) \rfloor$, and $j \mapsto x_j$ maps those with $h_j = h$ injectively into the divisors of $a + h$ in $[A, 2A)$. □

Definition 8.2. Let $G\colon \mathbb{N} \to \mathbb{R}$. We say that $\mathrm{SD}(G)$ holds if there are C, x_0 such that for all integers $x \geq x_0$, $x^{1/4} \leq y \leq x$ and $A \geq 1$,

$$\sum_{x < n \leq x+y} \tau_A(n) \leq C\, y\, G(x).$$

Lemma 8.3 (Geometric sums)**.** *Let $\alpha, \beta, s, t > 0$ and $K \geq 0$. Then*

$$\sum_{k<K} \min\bigl(\alpha 2^{-sk},\ \beta 2^{tk}\bigr) \leq \Bigl(\frac{1}{1-2^{-s}} + \frac{2^t}{2^t-1}\Bigr)\alpha^{t/(s+t)}\beta^{s/(s+t)}.$$

Proof. Let $M = \alpha^{t/(s+t)}\beta^{s/(s+t)}$. Then $(\alpha 2^{-sk})^t(\beta 2^{tk})^s = M^{s+t}$, so for each k either $\beta 2^{tk} \leq M$ or $\alpha 2^{-sk} < M$. The terms $\beta 2^{tk} \leq M$ grow with ratio 2^t, so they sum to at most $M2^t/(2^t-1)$. The terms $\alpha 2^{-sk} < M$ decay with ratio 2^{-s}, so they sum to at most $M/(1-2^{-s})$. □

Theorem 8.4. *Suppose $\mathrm{SD}(G)$ holds, $G \geq 1$, and $G(x) \ll x^{1/40}$. Then $P(a,b) \ll a^{3/10}\, G(a)^{1/5}$ uniformly for $0 < b < a$.*

Proof. Small a are handled by $P(a,b) \leq b < a$. Fix a dyadic window $[A, 2A)$, $A = 2^k$, with $T \geq 20$ visits. Proposition 7.3 gives $T \leq \alpha A^{-1/3}$, where $\alpha = (96 \cdot 10^6)^{1/6}a^{1/2}$. Take $Y = \max(\lfloor 2A/(T-1) \rfloor, \lceil a^{1/4} \rceil) \leq a$ in Proposition 8.1. With $H = \lfloor 2A/(T-1) \rfloor$, $\mathrm{SD}(G)$ gives $T \leq 2CG(a)(H + a^{1/4} + 1)$. Multiplying by $T-1$, using $H(T-1) \leq 2A$, and solving the quadratic gives $T \leq 1 + 2CG(a)(a^{1/4}+1) + 2\sqrt{CG(a)A}$, hence, as $T \geq 20$, $T \leq \gamma + \beta A^{1/2}$, where $\beta = 4\sqrt{CG(a)}$ and $\gamma = 4CG(a)(a^{1/4}+1)$. So every window satisfies

$$T_k \leq 19 + \gamma + \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr).$$

Lemma 8.3 with $s = 1/3$ and $t = 1/2$, where $1/(1-2^{-1/3}) + \sqrt{2}/(\sqrt{2}-1) < 9$, gives

$$\sum_k \min\bigl(\alpha\, 2^{-k/3},\ \beta\, 2^{k/2}\bigr) \leq 9\, \alpha^{3/5}\beta^{2/5} \ll a^{3/10}G(a)^{1/5},$$

with no factor $\log a$. The remaining terms contribute $(\log_2 a + 1)(19 + \gamma) \ll a^{1/100} \cdot a^{1/4}G(a)$. This is $\ll a^{3/10}G(a)^{1/5}$, because $G(a)^{4/5} \ll a^{1/50}$. □

The Erdős–Hooley function. Let $\Delta_{\mathrm{H}}(n) = \max_{u\in\mathbb{R}} \#\{d \mid n : \ e^u < d \le e^{u+1}\}$. Taking $e^{u+1} = 2A$, and using $e > 2$, gives $\tau_A(n) \le \Delta_{\mathrm{H}}(n)$. We use two published theorems.

(NT) Nair and Tenenbaum [9] prove a general upper bound for short sums of functions F satisfying $F(mn) \le \min(\lambda_1^{\Omega(m)}, \lambda_2 m^{\varepsilon})F(n)$ for $(m,n) = 1$. Since $\Delta_{\mathrm{H}}(mn) \le \tau(m)\Delta_{\mathrm{H}}(n)$ for $(m,n) = 1$, the function Δ_{H} satisfies this with $\lambda_1 = 2$ for every $\varepsilon > 0$. Their bound, for the polynomial X, gives for every fixed $\sigma > 0$, x large and $x^{\sigma} \le y \le x$

$$\sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll_{\sigma} \frac{y}{\log x} \sum_{n\le x} \frac{\Delta_{\mathrm{H}}(n)}{n}. \tag{9}$$

We use it with $\sigma = 1/4$. In the parameters of their Theorem 1, the polynomial degree is $g = 1$; take $\varepsilon = 1/16$ and $\delta = 1$. Then $4g^2\varepsilon = 1/4$ and the required class is $\mathcal{M}_1(2, B, 1/48)$ for a suitable B. The same consequence is displayed immediately after equation (2) of their paper.

(BT) de la Bretèche and Tenenbaum [2, Theorem 1.1] prove that for large x

$$\sum_{n\le x} \Delta_{\mathrm{H}}(n) \ll x\,(\log\log x)^{5/2}. \tag{10}$$

Lemma 8.5. *Assume* (10)*, and let* $\ell(x) = \max(1, \log\log x)$*. Then* $\sum_{n\le x} \Delta_{\mathrm{H}}(n)/n \ll \log x \cdot \ell(x)^{5/2}$ *for large* x*.*

Proof. Split n into dyadic blocks $2^j \le n < 2^{j+1} \le 2x$. The terms with n below the threshold of (10) contribute $O(1)$, since $\Delta_{\mathrm{H}}(n) \le n$. Each remaining block contributes at most $2^{-j}\sum_{n\le 2^{j+1}} \Delta_{\mathrm{H}}(n) \ll \ell(2x)^{5/2}$, and $\ell(2x) \le 2\ell(x)$. There are at most $\log_2 x + 1 \le 3\log x$ blocks. □

Proof of Theorem 1.3. By (9) and Lemma 8.5, $\sum_{x<n\le x+y} \tau_A(n) \le \sum_{x<n\le x+y} \Delta_{\mathrm{H}}(n) \ll y\,\ell(x)^{5/2}$ for $x^{1/4} \le y \le x$. So SD(G) holds with $G = \ell^{5/2}$, which satisfies $G \ge 1$ and $G(x) \ll x^{1/40}$. Theorem 8.4 gives $P(a,b) \ll a^{3/10}\ell(a)^{1/2} \ll a^{3/10}\sqrt{\log\log 3a}$. □

9. Rises in the quotient increments

Throughout this section $k_i = q_{i+1} - q_i \ge 1$, $d_i = b_i - b_{i+1} \ge 1$, and a *rise* is an index $0 \le i \le P(a,b) - 3$ with $k_i < k_{i+1}$.

Lemma 9.1 (Six quotients in arithmetic progression)**.** *Let* $q, h \ge 1$ *and* $x_0 > x_1 > \cdots > x_6 \ge 0$ *satisfy* $x_j(q + jh) + x_{j+1} = a$ *for* $0 \le j \le 5$*. Then* $q^4 \le 324\,h^3 a$*.*

Proof. Let $f_j = x_j - x_{j+1}$, and let $E = (f_1 - f_2)h$ and $F = (f_2 - f_3)h$ be the curvatures of the blocks starting at x_1 and x_2 (all their increments equal h). With $U = f_0 - 2f_1 + f_2$, the division equations $\mathcal{E}_j\colon x_j(q + jh) + x_{j+1} - a = 0$ give the exact identity

$$qhU = (3h-1)E + F, \qquad \text{namely} \qquad qhU - (3h-1)E - F = h(\mathcal{E}_0 - 3\mathcal{E}_1 + 3\mathcal{E}_2 - \mathcal{E}_3).$$

This is the case $u = v = w = z = h$ of Lemma 7.1, where $L = 2h^3U$. By Lemma 3.2, $E, F \ge 1$, so the integer U is at least 1.

Now bound E and F. Subtracting consecutive division equations gives $q_j f_j \le h x_{j+1}$, where $q_j = q + jh$. Since $x_{j+1}q_{j+1} \le a$, this yields $q^2 f_j \le ha$ for $1 \le j \le 4$. By Lemma 3.4, $Ex_3 \le (h+1)(f_1+f_2+f_3)^2 \le 9h^2(h+1)a^2/q^4$. Also $a < x_3(q+3h+1)$, so $Eq^4 \le 9h^2(h+1)(q+3h+1)a$, and likewise $Fq^4 \le 9h^2(h+1)(q+4h+1)a$. Hence

$$q^5 h \le q^4\big((3h-1)E + F\big) \le 27h^3(h+1)(q+4h+1)a.$$

If $q \ge h$ this gives $q^4 \le 324h^3a$. If $q < h$ then $q^4 < h^4 \le h^3 a$, since $h \le q_1 \le a$. □

Lemma 9.2. *Whenever $b_{i+2} > 0$, $d_{i+1}k_i < d_i k_{i+1}$. In particular, if $k_{i+1} \le k_i$ then $d_{i+1} < d_i$.*

Proof. This is $D = d_i k_{i+1} - d_{i+1}k_i \ge 1$ (Lemma 3.2). If $k_{i+1} \le k_i$ and $d_{i+1} \ge d_i$, then $d_i k_{i+1} \le d_{i+1}k_i$, which is a contradiction. □

Lemma 9.3. *Let $c_j \ge 1$ be integers, $j \in J$, such that no value occurs more than M times. Then $|J|^2 \le 2M \sum_{j\in J} c_j$.*

Proof. Induct on $|J|$. Let c_{j_0} be the largest value. The values lie in $[1, c_{j_0}]$, so $|J| \le M c_{j_0}$. Removing j_0 and using $|J|^2 = (|J|-1)^2 + 2|J| - 1$ completes the induction. □

Proposition 9.4. *Let $x_0, \dots, x_{T-1}$ be a window run in $[A, 2A)$ with $T \ge 2$, and let ρ be the number of rises among its $T-1$ increments. Then:*

1. *$(T-1)^2 \le 2(\rho+1)(A-1)$;*
2. *$T - 1 \le \lfloor a/A \rfloor$;*
3. *if $6A \le a$, then $T - 1 = \ell + s$ with $\ell^2 \le 8(\rho+1)\lfloor a/A \rfloor$ and $s^3 \le 26244\, A$.*

Proof. Split the increments into *runs*: maximal stretches containing no rise. There are at most $\rho + 1$ runs.

(1) By Lemma 9.2, the drops strictly decrease along a run, so each value of d_j occurs at most once per run, hence at most $\rho + 1$ times. Lemma 9.3 and $\sum_j d_j = x_0 - x_{T-1} \le A - 1$ give the claim.

(2) The increments are ≥ 1, and they sum to $q_{T-1} - q_0 \le \lfloor a/A \rfloor$.

(3) Call an increment h *low* if $26244h^3A^4 < a^3$. Five consecutive equal increments h inside the run give six quotients in arithmetic progression. Lemma 9.1 then gives $q^4 \le 324h^3a$. On the other hand, $a < 3Aq$, which follows from $a = qx_0 + x_1$, $x_1 < x_0 < 2A$ and $6A \le a$. Together these give $a^3 \le 26244h^3A^4$, so h is not low. Within a run the increments are nonincreasing, so equal values are consecutive. Hence each low value occurs at most 4 times per run, and at most $4(\rho+1)$ times in all. Lemma 9.3 bounds the number ℓ of low increments by $\ell^2 \le 8(\rho+1)\sum_j k_j \le 8(\rho+1)\lfloor a/A \rfloor$. Let s be the number of high increments. If $s = 0$, its bound is immediate. Otherwise let h_* be their smallest value. Then $sh_* \le \lfloor a/A \rfloor$ and $a^3 \le 26244h_*^3A^4$, so $s^3a^3 \le 26244A^4(a/A)^3 \le 26244Aa^3$. □

Proof of Theorem 1.4. The rises inside a window are rises of the orbit, so $\rho + 1 \le R$. Combining the three parts of Proposition 9.4, as in the proof of Theorem 8.4, gives for the window $[2^k, 2^{k+1})$ (when $6A > a$, part (2) gives $T - 1 \le a/A < 6 \le 30A^{1/3}$)

$$T_k \le 1 + \sqrt{R}\,\min\bigl(\sqrt{8a}\,2^{-k/2},\ \sqrt{2}\,2^{k/2}\bigr) + \min\bigl(a\,2^{-k},\ 30\cdot 2^{k/3}\bigr).$$

Lemma 8.3 gives $\sum_k \min(\sqrt{8a}\,2^{-k/2}, \sqrt{2}\,2^{k/2}) \ll (\sqrt{8a}\sqrt{2})^{1/2} = 2a^{1/4}$. It also gives $\sum_k \min(a2^{-k}, 30\cdot 2^{k/3}) \ll a^{1/4}$. Since $\log_2 a + 1 \ll a^{1/4}$, summing over k gives the theorem. □

Remark 9.5. By Theorem 1.4, an orbit with $P(a,b) \asymp a^{3/10}$ must have $R \gg a^{1/10}$. Proposition 9.4(3) says more: inside the critical window $A \asymp a^{3/5}$, a run of $T \asymp a^{3/10}$ remainders must contain $\gg a^{1/5}$ rises.

10. The Hankel hierarchy

In this section and the next five we write $N = P(a,b)$, so that $b_0 > \cdots > b_{N-1} > 0 = b_N$, and $k_i = q_{i+1} - q_i$. For each remaining tail we construct a probability measure on its quotient differences. Reweighting by the reciprocal quotient will make its total mass equal to b_i/a.

Put $p_i(j) = q_{i+j} - q_i$ for $0 \le j < N - i$; the symbols p_i, w_i are local to this construction. Define the weights backwards from $w_{N-1}(0) = 1$ by

$$w_i(0) = 1 - \sum_{j=0}^{N-i-2} \frac{w_{i+1}(j)}{p_i(j+1)}, \qquad w_i(j+1) = \frac{w_{i+1}(j)}{p_i(j+1)} \quad (0 \le j < N-i-1).$$

Let μ_i be the measure with mass $w_i(j)$ at $p_i(j)$, and let

$$d\nu_i(x) = \frac{d\mu_i(x)}{q_i + x}, \qquad h_i(s) = \int x^s \, d\nu_i = \sum_{j=0}^{N-i-1} \frac{w_i(j) p_i(j)^s}{q_{i+j}}.$$

The recursion is chosen to reproduce the division equation $q_i h_i(0) + h_{i+1}(0) = 1$ for $i+1 < N$.

Lemma 10.1. (1) *The weights are positive and sum to* 1.
(2) $h_i(0) = b_i/a$, *that is,* $b_i/a = \mathbb{E}\big[1/(q_i + X_i)\big]$ *for* $X_i \sim \mu_i$; *and* $h_{N-1}(s) = 0$ *for* $s \ge 1$.
(3) $h_i(s+1) = \sum_{t \le s} \binom{s}{t} k_i^{s-t} h_{i+1}(t)$ *for* $i + 1 < N$.
(4) $h_i(s) = (-q_i)^s b_i/a + z$ *with* $z \in \mathbb{Z}$.

Proof. (1) Since $p_i(j+1) \ge 1$, the sum defining $w_i(0)$ is at most 1. It is strictly less: if $N - i - 1 \ge 2$, the term $j = 1$ has $p_i(2) \ge 2$; if $i = N - 2$, the last gap satisfies $q_{N-1} \ge q_{N-2} + 2$, because $b_{N-1} \mid a$ gives $q_{N-1} - q_{N-2} - 1 = q_{N-2}(b_{N-2} - b_{N-1})/b_{N-1} \ge 1$. (2) Since $p_i(j+1) = q_{i+j+1} - q_i$, the definitions give $q_i h_i(0) + h_{i+1}(0) = 1$, which is the normalised division equation; and $h_{N-1}(0) = 1/q_{N-1} = b_{N-1}/a$; since $\mu_{N-1} = \delta_0$, also $h_{N-1}(s) = 0$ for $s \ge 1$. (3) Use $p_i(j+1) = k_i + p_{i+1}(j)$ and the binomial theorem. (4) For $i = N - 1$ this holds with $z = -(-q_{N-1})^s b_{N-1}/a = -(-1)^s q_{N-1}^{s-1}$ for $s \ge 1$, since $q_{N-1} b_{N-1} = a$. Then induct backwards on i using (3), $b_{i+1}/a = 1 - q_i b_i/a$ and $k_i - q_{i+1} = -q_i$. □

Theorem 10.2. *For* $1 \le r \le N - i$, *the number* $I_r(i) = a \det\big(h_i(j+l)\big)_{0 \le j,l < r}$ *is a positive integer. Moreover* $I_1(i) = b_i$, $I_2(i) = d_i$ *(if* $i+1 < N$*),* $I_3(i) = D_i$ *(if* $i+3 < N$*), and* $I_4(i) = L_i$ *(if* $i+5 < N$*), where* D_i *and* L_i *are the curvature and the six-step integer of the blocks starting at* b_i.

Proof. Integrality. By Lemma 10.1(4), $h_i(s+1) + q_i h_i(s) \in \mathbb{Z}$. Multiply the Hankel matrix on the left by the unit lower triangular matrix that adds q_i times row $j-1$ to row j. This makes rows $1, \dots, r-1$ integral and leaves the determinant unchanged. Expand along row 0, whose entries are $(-q_i)^l b_i/a + (\text{integer})$; then $a \det$ is an integer.

Positivity. The Hankel matrix is the Gram matrix of $1, x, \dots, x^{r-1}$ in $L^2(\nu_i)$. Its quadratic form is $\sum_j w_i(j) q_{i+j}^{-1} \big(\sum_l c_l p_i(j)^l\big)^2$. If this vanishes, the polynomial $\sum_l c_l x^l$ vanishes at the $N - i \ge r$ distinct points $p_i(j)$, so it is zero, by the invertibility of the Vandermonde matrix. So the matrix is positive definite.

The first four. First, $I_1(i) = a h_i(0) = b_i$. For $i+1 < N$, write $\beta_j = b_{i+j}/a$. Then $h_i(0) = \beta_0$, $h_i(1) = \beta_1$ and $h_i(2) = k_i \beta_1 + \beta_2$. Hence

$$I_2(i) = a\big(\beta_0(k_i \beta_1 + \beta_2) - \beta_1^2\big) = a(\beta_0 - \beta_1) = d_i,$$

using $q_i \beta_0 + \beta_1 = q_{i+1} \beta_1 + \beta_2 = 1$. For the larger determinants, Lemma 10.1(2),(3) makes the entries $h_i(s)$, $s \le 2r - 2$, explicit polynomials in $\beta_j = b_{i+j}/a$ and $k_i, \dots, k_{i+2r-4}$; at the end of the orbit this uses $h_{N-1}(s) = 0$ for $s \ge 1$, that is, $b_N = 0$. Expanding and reducing by $q_{i+j} \beta_j + \beta_{j+1} = 1$ gives D_i and L_i in the stated ranges. The formal proof records polynomial certificates for these two reductions. □

The ranges $i + 3 < N$ and $i + 5 < N$ cannot be relaxed to all i at which D_i or L_i is defined. For $a = 926$ and $b = 229$ the orbit is $229, 10, 6, 2$, so $N = 4$, and $I_3(1) = 984$ while $D_1 = 988$.

Remark 10.3. In these index ranges, the positivity of D and L, proved earlier by explicit identities, is an instance of the positivity of Gram determinants. The integrality of L (Lemma 7.1(1)) is an instance of the integrality of I_4. A priori $I_r(i)$ involves the moments $h_i(s)$ for $s \le 2r-2$, and hence $b_i, \dots, b_{i+2r-2}$. In fact $I_3(i)$ depends only on b_i, b_{i+1}, b_{i+2} and q_i, q_{i+1}, q_{i+2}. Similarly $I_4(i)$ depends only on $b_i, \dots, b_{i+3}$ and $q_i, \dots, q_{i+4}$: the later remainders cancel.

11. Breaking the exponent 3/10

In this section we prove Theorem 1.5. We keep the notation of Section 10: $b_0 > b_1 > \cdots > b_{N-1} > 0$ is the positive part of the orbit, $N = P(a,b)$, and $q_i = \lfloor a/b_i \rfloor$.

11.1. **Heuristics.** The exponent 3/10 of Theorem 1.2 comes from a single balanced configuration: $T \asymp a^{3/10}$ remainders in a window $[A, 2A)$ with $A \asymp a^{3/5}$, quotients $q \asymp a^{2/5}$ and typical increments $k \asymp a^{1/10}$. At this scale the four-term Hankel invariant $I_4(i) = L(i)$ has size $\asymp aH^6/q^4 \asymp 1$, where H bounds the increments. The idea is that a positive integer of bounded size carries information. The labels $I_4(i), I_4(i+1), I_4(i+2)$, together with four further integers of the same size, determine a *pattern*. States with a given pattern satisfy two linear equations and one quadratic equation in five consecutive increments. So a pattern accounts for at most $O(H^2)$ increment tuples, and each tuple for at most four states. At the balanced point this gives $T \ll H^2 \asymp a^{1/5}$, a large saving. Moving away from the balanced point costs a power of the pattern size, and optimising gives the exponent 23/77.

11.2. **The pattern matrix.** Fix a state i with $i + 6 < N$ and write, for $m \ge 0$,

$$k_m = q_{i+m+1} - q_{i+m}, \qquad d_m = b_{i+m} - b_{i+m+1},$$
$$s_m = k_0 + \cdots + k_{m-1}, \qquad r_m = k_m(s_m + s_{m+1}) + k_{m+1}.$$

All indices here are local to the chosen state, and the definitions apply whenever the indicated orbit entries exist; the empty sum gives $s_0 = 0$. So $s_m = q_{i+m} - q_i$, $r_0 = k_0^2 + k_1$, $r_1 = k_1(2k_0 + k_1) + k_2$, and so on. Consider the 3×4 integer matrix

$$\mathcal{M}_i = \begin{pmatrix} d_0 & d_1 & d_2 & d_3 \\ k_0 & k_1 & k_2 & k_3 \\ r_0 & r_1 & r_2 & r_3 \end{pmatrix},$$

and let $U_m(i) = (-1)^{m+1} \det(\mathcal{M}_i \text{ with column } m \text{ deleted})$ for $0 \le m \le 3$.

Lemma 11.1. (1) *$U_3(i) = L(i) = I_4(i)$ and $U_0(i) = -L(i+1) = -I_4(i+1)$, where L is the six-step integer of Lemma 7.1.*
(2) *$\sum_m U_m(i)\, d_m = \sum_m U_m(i)\, k_m = \sum_m U_m(i)\, r_m = 0$.*

Proof. (2) is the Laplace expansion of a 4×4 determinant with a repeated row. For (1), the identity $\det(\text{columns } 0,1,2) = L(i)$ holds as a polynomial identity in $d_0, d_1, d_2, k_0, \dots, k_3$, by direct expansion; and $L(i) = I_4(i)$ by Theorem 10.2. Passing from i to $i+1$ shifts the d- and k-rows by one column, and, since $s_m(i+1) = s_{m+1}(i) - k_0$, it replaces r_m by $r_{m+1} - 2k_0k_{m+1}$. This row operation does not change the determinant, so $\det(\text{columns } 1,2,3) = L(i+1)$. □

Remark 11.2. The vector $C_m = U_m/U_3$ has a meaning: $\pi_i = \sum_{m \le 3} C_m(F_{m+1} - F_m)$, with Newton polynomials $F_m(x) = \prod_{l<m}(x - s_l)$, is the monic orthogonal polynomial of degree four for the measure ν_i of Section 10. The three rows of $\mathcal{M}_i$ encode $\nu_i(\pi_i) = 0$ and $\mu_i(x\pi_i) = \mu_i(x^2\pi_i) = 0$, and $\nu_i(xf) = \mu_i(f) - q_i\nu_i(f)$ turns these into $\nu_i(x^j\pi_i) = 0$ for $j \le 3$. In particular $U_m(i)/I_4(i)$ are tail sums of the Newton coefficients of π_i: writing $\pi_i = \sum_{l \le 4} \gamma_l F_l$, one has $U_m(i)/I_4(i) = \gamma_{m+1} + \cdots + \gamma_4$, and $\gamma_0 = -C_0 = I_4(i+1)/I_4(i)$. This interpretation is not needed for the elementary count in this section; it will be used in Section 15.

The *pattern* of i is the vector of seven integers

$$\theta(i) = \big(L(i),\, L(i+1),\, L(i+2),\, U_1(i),\, U_2(i),\, U_1(i+1),\, U_2(i+1)\big),$$

and its *gaps* are $\kappa(i) = (k_0, \dots, k_4)$. By Lemma 11.1, $\theta(i)$ determines $U(i)$ and $U(i+1)$ completely.

11.3. **Three consequences.**

Lemma 11.3 (The reduced quadratic)**.** *Write $\theta(i) = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ and $\kappa(i) = (k_0, \dots, k_4)$. Then*

$$N_1 N_0 (N_1 + N_0)\, k_3^2 + \beta\, k_3 + \gamma = 0,$$

where β and γ are polynomials in $\theta(i)$, k_1 and k_2 alone.

Proof. Lemma 11.1(2) for the k-row at i and at $i+1$ gives

$$N_1 k_0 = u_1 k_1 + u_2 k_2 + N_0 k_3, \qquad N_1 k_4 = N_2 k_1 - v_1 k_2 - v_2 k_3.$$

Substitute these into $N_1^2 \sum_m U_m(i) r_m = 0$. Since k_0 enters the r_m at most quadratically and k_4 only linearly, the result is a polynomial in (θ, k_1, k_2, k_3). The coefficient of k_3^2 comes from $-N_1 \cdot N_1^2 k_0^2$, from $N_0 \cdot N_1^2 \cdot 2k_0 k_3$ and from $N_0 \cdot N_1^2 k_3^2$, and it equals $-N_1 N_0^2 + 2N_1 N_0^2 + N_1^2 N_0 = N_1 N_0 (N_1 + N_0)$. Explicitly

$$\beta = 2N_1^2 N_0 (k_1 + k_2) + N_1^2 u_2 + 2N_1 N_0 (u_1 k_1 + u_2 k_2) - N_1 N_0 v_2,$$

$$\gamma = N_1 (u_1 k_1 + u_2 k_2)^2 + N_1^2 \big(u_1 k_1^2 + 2u_2 k_1 k_2 + u_2 k_2^2 + u_1 k_2 - N_1 k_1\big) + N_0 N_1 (N_2 k_1 - v_1 k_2).$$

□

Lemma 11.4 (The multiplicity identity)**.** *Let $F_m(Y) = \prod_{l<m} (Y - s_l)$ and $\mathcal{Q}_i(Y) = \sum_{m=0}^{3} U_m(i)\, F_m(Y)\, (Y - s_m - 1)$. Then*

$$\mathcal{Q}_i(-q_i) = a\, H_4(k_0, \dots, k_4),$$

where H_4 is an explicit polynomial with 43 terms, independent of i. As a polynomial in q, $\mathcal{Q}_i(-q)$ has degree four and leading coefficient $U_3(i) = L(i) \geq 1$.

Proof. The division equations read $b_{i+m+1} = a - (q_i + s_m) b_{i+m}$. Use them to eliminate $b_{i+1}, \dots, b_{i+4}$ from both sides. The difference becomes the zero polynomial in $a, q_i, b_i, k_0, \dots, k_4$; explicit certificates are in the Lean proof. The polynomial H_4 is the Hankel determinant $\det\big(\mu_i(x^{j+l})\big)_{j,l<4}$, and the identity is $I_4(i)\, \pi_i(-q_i) = a \det\big(\mu_i(x^{j+l})\big)$. This follows from the determinant formula for π_i, since $\nu_i(x^{s+1}) + q_i \nu_i(x^s) = \mu_i(x^s)$. The leading coefficient comes from the term $m = 3$. □

Lemma 11.5 (Labels)**.** *Let $i + 6 < N$ and put*

$$G(k_0, \dots, k_4) = (k_0 + 1)\big(k_1 s_2 + s_2 + k_1 + k_2 + 1\big)\big(k_2 (k_1 + k_2) s_3 + e_3\big),$$

where $e_3 = s_3(k_1 + k_2) + s_3 k_2 + (k_1 + k_2) k_2 + (k_0 + 2k_1 + 3k_2)(k_3 + 1) + k_3^2 + 2k_3 + k_4 + 1$. Then $L(i)\, q_i^4 \leq a\, G(k_0, \dots, k_4)$. If moreover $b_i < 2A$, then $L(i)\, a^3 \leq 256 A^4\, G(k_0, \dots, k_4)$.

Proof. The Newton polynomials $F_0, \dots, F_3$ are monic of degrees $0, \dots, 3$. So the matrix $\big(\nu_i(F_j F_l)\big)$ is $\mathcal{U}\mathcal{H}\mathcal{U}^{\mathsf{T}}$, where $\mathcal{H}$ is the Hankel matrix of ν_i and $\mathcal{U}$ is unit lower triangular. Hence it has determinant $I_4(i)/a$, and it is positive definite, as in the proof of Theorem 10.2. By Hadamard's inequality,

$$\frac{I_4(i)}{a} \leq \prod_{j<4} \nu_i(F_j^2) \leq \prod_{j<4} \frac{\mu_i(F_j^2)}{q_i},$$

since $\nu_i = \mu_i/(q_i + x) \le \mu_i/q_i$ on the nonnegative support. By the shift rule $\mu_i(F_j g) = \mu_{i+j}\big(g(x + s_j)\big)$ and the moments $\mu_{i+m}(1) = \mu_{i+m}(x) = 1$, $\mu_{i+m}(x^2) = k_m + 1$, $\mu_{i+m}(x^3) = k_m^2 + 2k_m + k_{m+1} + 1$, the product equals G. For the second claim, $a = q_i b_i + b_{i+1} < 2A(q_i + 1) \le 4Aq_i$, so $L(i)a^4 \le 256A^4 L(i) q_i^4 \le 256A^4 aG$. □

11.4. **The size of a pattern.**

Lemma 11.6. *Let $1 \le k_0, \dots, k_5 \le H$ and $A \ge 1$. Suppose $N_0 = L(i)$ and $N_1 = L(i+1)$ satisfy $N_0 a^3 \le 256A^4 G(k_0, \dots, k_4)$ and $N_1 a^3 \le 256A^4 G(k_1, \dots, k_5)$, and that the* scale condition

$$237568\, A^4 H^5 \le a^3 \tag{11}$$

holds. Then $L(i)$, $L(i+1)$, $|U_1(i)|$ and $|U_2(i)|$ are all at most $172032\, A^4 H^6/a^3$.

Proof. Put $V = k_1 k_2 (k_1 + k_2)$ and $\Delta_4 = k_0 \cdot k_1 s_2 \cdot k_2 (k_1 + k_2) s_3$. Expanding G and bounding each of the three error terms by $56H^5$, $170H^5$ and $238H^5$ gives $G(k_0, \dots, k_4) \le \Delta_4 + 464H^5$. Also $\Delta_4 \le 6H^3 V$, because $\Delta_4 = k_0 s_2 s_3 \cdot V$. Similarly $G(k_1, \dots, k_5) \le 6H^3 V + 464H^5$, because the corresponding product is $V \cdot k_3(k_2 + k_3)(k_1 + k_2 + k_3)$. By (11) and $N_0, N_1 \ge 1$, $256A^4 \cdot 464H^5 \le a^3/2 \le N_j a^3/2$ for $j = 0, 1$, so

$$N_0 a^3,\ N_1 a^3 \le 3072\, A^4 H^3 V.$$

Since $N_0 \ge 1$, this and (11) give $V \ge 77H^2$.

Now solve for $(u_1, u_2) = (U_1(i), U_2(i))$ from the k- and r-relations of Lemma 11.1. These read $u_1 k_1 + u_2 k_2 = N_1 k_0 - N_0 k_3$ and $u_1 r_1 + u_2 r_2 = N_1 r_0 - N_0 r_3$. The determinant is

$$\delta = k_1 r_2 - k_2 r_1 = k_1 k_2 (k_1 + k_2) + k_1 k_3 - k_2^2 \ \ge\ V - H^2 \ \ge\ V/2.$$

Using $r_m \le (2m + 2)H^2$, Cramer's rule gives $|u_j|\delta \le 14(N_0 + N_1)H^3$, hence

$$|u_j|\, V a^3 \le 28H^3 (N_0 + N_1) a^3 \le 28H^3 \cdot 6144 A^4 H^3 V,$$

that is, $|u_j| a^3 \le 172032 A^4 H^6$. Finally $N_j a^3 \le 3072 A^4 H^3 V \le 6144 A^4 H^6$, since $V \le 2H^3$. □

The point of the argument is in the first paragraph. The two adjacent labels share the Vandermonde factor V, so the ratio $L(i+1)/L(i)$ never enters the bound.

11.5. **The counting lemma.**

Proposition 11.7. *Let $\mathcal{G}$ be a set of states i with $i + 8 < N$. Suppose that for each $i \in \mathcal{G}$ all entries of $\theta(i)$ have absolute value at most M, and all entries of $\kappa(i)$ lie in $[1, H]$. Then $|\mathcal{G}| \le 8(2M + 1)^7 H^2$.*

Proof. (i) At most four states per (θ, κ). If $\theta(i) = \theta$ and $\kappa(i) = \kappa$, then by Lemma 11.4 q_i is a root of the polynomial $\mathcal{Q}(-q) - aH_4(\kappa)$. This polynomial depends only on (θ, κ), and it has degree four and leading coefficient $\theta_0 \ge 1$. Since q_i is strictly increasing in i, there are at most four such states.

(ii) At most two values of κ per (θ, k_1, k_2). By Lemma 11.3, k_3 is a root of a quadratic whose coefficients depend only on (θ, k_1, k_2). The leading coefficient is $N_1 N_0 (N_1 + N_0) > 0$. The two linear relations then determine k_0 and k_4, because $N_1 \ne 0$.

(iii) There are at most $(2M + 1)^7 H^2$ values of (θ, k_1, k_2). □

11.6. **The window bound.**

Proposition 11.8. *Let $b_s, \dots, b_{s+T-1}$ be a window run in $[A, 2A)$ with $T \geq 36$, and put $H = \lfloor 28a/(AT) \rfloor + 1$ and $M = \lfloor 172032 A^4 H^6 / a^3 \rfloor$. If (11) holds, then*

$$T \leq 16\,(2M+1)^7 H^2.$$

Proof. Call a start i *good* if $s \leq i$, $i + 9 \leq s + T$ and $q_{j+1} - q_j \leq H$ for $i \leq j \leq i + 6$, that is, $k_0, \dots, k_6 \leq H$ at state i.

Most starts are good. Let $\mathcal{B}$ be the set of $j \in [s, s+T-1)$ with $q_{j+1} - q_j > H$. Then $|\mathcal{B}|(H+1) \leq q_{s+T-1} - q_s \leq a/A$. Since $H \cdot AT > 28a$, this gives $28|\mathcal{B}| \leq T$. Each $j \in \mathcal{B}$ spoils at most seven starts, so at least $T - 9 - 7|\mathcal{B}| \geq T/2$ starts are good, as $T \geq 36$.

Good starts satisfy Proposition 11.7. Let i be good. Then $i+8 < N$, and $b_i, b_{i+1}, b_{i+2} < 2A$. So Lemma 11.5 applies at i, $i+1$ and $i+2$. Lemma 11.6 at i and at $i+1$ bounds all seven entries of $\theta(i)$ by M. Therefore $T/2 \leq 8(2M+1)^7 H^2$. □

11.7. **Proof of Theorem 1.5.**

Proposition 11.9. *Let $T \geq 36$ be the length of a window run in $[A, 2A)$. Then $T^{77} \leq C_{77} \Delta^{22} a^{23}$ for an absolute constant C_{77}.*

Proof. We prove this with

$$C_{77} = 122449^{20} + (8 \cdot 237568 \cdot 56^5)^{11} + (64 \cdot 50176)^{11} + 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}.$$

We use Proposition 7.3, $T^6 A^2 \leq 96 \cdot 10^6 a^3$, and Proposition 5.1, which gives $T^2 \leq 2T(T-1) \leq 8A\Delta$.

Case $AT > 28a$. Then $784 a^2 T^4 \leq T^6 A^2 \leq 96 \cdot 10^6 a^3$, so $T^4 \leq 122449\, a$ and $T^{77} \leq 122449^{20} a^{20}$.

From now on $AT \leq 28a$, so $H \cdot AT \leq 28a + AT \leq 56a$.

Case (11) fails. Then $a^3 (AT)^5 < 237568 A^4 (H\,AT)^5 \leq 237568 \cdot 56^5 A^4 a^5$, so $AT^5 \leq 237568 \cdot 56^5 a^2$. Hence $T^7 = T^5 T^2 \leq 8\Delta\, AT^5 \leq 8 \cdot 237568 \cdot 56^5\, \Delta a^2$. Raise this to the eleventh power.

Case (11) holds and $M = 0$. Then $T \leq 16H^2$, so $T(AT)^2 \leq 16(56a)^2$. Hence $T^7 = T^3 (T^2)^2 \leq 64 \Delta^2\, T^3 A^2 \leq 64 \cdot 50176\, \Delta^2 a^2$. This case is in fact vacuous: the proof of Proposition 11.8 provides a good start i, and there $1 \leq L(i) \leq M$ by Lemma 11.6. We treat it anyway, here and in the analogous proofs below, so that the constants match the formal proof.

Case (11) holds and $M \geq 1$. Then $2M + 1 \leq 3M$ and $Ma^3 \leq 172032 A^4 H^6$, so

$$T\, a^{21} \leq 16 \cdot 3^7\, (Ma^3)^7 H^2 \leq 16 \cdot 3^7 \cdot 172032^7\, A^{28} H^{44}.$$

Multiplying by $(AT)^{44}$ and using $H \cdot AT \leq 56a$ gives $T^{45} A^{16} \leq 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\, a^{23}$. Finally

$$T^{77} = T^{45} (T^2)^{16} \leq 8^{16} \Delta^{16}\, T^{45} A^{16} \leq 8^{16} \cdot 16 \cdot 3^7 \cdot 172032^7 \cdot 56^{44}\, \Delta^{16} a^{23}.$$

In every case $T^{77} \leq C_{77} \Delta^{22} a^{23}$. □

Proof of Theorem 1.5. This is the argument of Theorem 1.2, with Proposition 11.9 in place of Corollary 5.3. Let $\varepsilon > 0$ and choose an integer $m > 1/\varepsilon$. By Lemma 2.2, $\Delta^{22m} \leq C_{22m} \cdot 32 a^7$, so every dyadic window has $T \leq 35$ or

$$T^{77m} \leq C_{77}^m\, C_{22m}\, 32\; a^{23m+7}.$$

Since $(23m+7)/(77m) = 23/77 + 1/(11m) \leq 23/77 + \varepsilon/2$, each of the $\lfloor \log_2 a \rfloor + 1$ windows has $T \ll_\varepsilon a^{23/77 + \varepsilon/2}$, and the theorem follows. □

Remark 11.10. In terms of the scales, Proposition 11.8 says $T \ll X^7 H^2$, where $X = A^4 H^6/a^3$ bounds the seven pattern integers and $H \asymp a/(AT)$. Thus $X \asymp a^3/(A^2T^6)$ and $T^{45}A^{16} \ll a^{23}$. The drop bound $A \gg T^2/\Delta$ then gives $T^{77} \ll \Delta^{16}a^{23}$. The scale condition (11) says $Q^4 \gg aH^5$, where $B = a/A$, and its failure gives the stronger bound $T \ll a^{2/7+o(1)}$.

12. The ternary quadratic and the exponent 11/37

Proposition 11.7 uses only one coefficient of the reduced quadratic. Here we use all of it, which saves a factor H in the count per pattern and proves Theorem 1.6. We keep the notation of Section 11, and, as in Lemma 11.3, write a pattern as $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2) = (L(i), L(i+1), L(i+2), U_1(i), U_2(i), U_1(i+1), U_2(i+1))$. In particular A is still the scale of the window $[A, 2A)$.

12.1. **Signs.** Throughout this subsection the pattern and gaps arise from the same state, so the cofactor relations of Lemma 11.1 remain in force. Put $S(a,b,c) = a^2b + (a-1)b^2 + ac$ and $E(a,b,c,d) = a^2c + (2a-1)bc + ac^2 + ad$. Both are positive at positive integers.

Lemma 12.1 (Signs)**.** *If $N_0, N_1 \geq 1$ and $k_0, \dots, k_4 \geq 1$, then $u_1 > 0 > u_2$.*

Proof. Solving the k- and r-relations of Lemma 11.1 for (u_1, u_2) by Cramer's rule gives the exact identities

$$u_1 S(k_1,k_2,k_3) = N_1 E(k_0,k_1,k_2,k_3) + N_0 S(k_2,k_3,k_4),$$
$$-u_2 S(k_1,k_2,k_3) = N_1 S(k_0,k_1,k_2) + N_0 E(k_1,k_2,k_3,k_4),$$

since the Cramer determinant is $k_1 r_2 - k_2 r_1 = S(k_1,k_2,k_3)$. □

12.2. **The ternary relation.** For a pattern θ let

$$F_\theta(x,y,z) = (u_1x + u_2y + N_0z)^2 + N_1u_1x^2 + N_1u_2y^2 + N_0N_1z^2 + 2N_1u_2xy + 2N_0N_1xz$$
$$+ 2N_0N_1yz + (N_0N_2 - N_1^2)x + (N_1u_1 - N_0v_1)y + (N_1u_2 - N_0v_2)z.$$

Lemma 12.2. *At every state i with $i+7 < N$, $F_{\theta(i)}(k_1,k_2,k_3) = 0$. Write $F_\theta = \alpha x^2 + \beta xy + \gamma y^2 + d(z)x + e(z)y + f(z)$. Then $\alpha = u_1(u_1+N_1)$, $\beta^2 - 4\alpha\gamma = -4N_1u_2(N_1+u_1)(u_1-u_2)$, and*

$$(\beta d_1 - 2\alpha e_1)^2 + (\beta^2 - 4\alpha\gamma)(4\alpha f_2 - d_1^2) = 16\, N_0N_1^2u_1(N_0-u_2)(N_1+u_1)^2(u_1-u_2),$$

where d_1, e_1, f_2 are the coefficients of z, z, z^2 in d, e, f, respectively. In particular, under the hypotheses of Lemma 12.1, $\alpha > 0$, $\beta^2 - 4\alpha\gamma > 0$ and the last quantity is nonzero.

Proof. The identity $N_1F_\theta(k_1,k_2,k_3) = N_1^2R + (N_0N_1k_3 - N_1^2k_0 + N_1u_1k_1 + N_1u_2k_2)K_0 - N_0N_1K_1$, where K_0, K_1 and R are the k-relations at i, $i+1$ and the r-relation at i, gives the first claim. The rest is expansion. So $F_\theta \equiv N_1R$ modulo the two linear relations. □

12.3. **Norm equations.**

Lemma 12.3 (Norm equations)**.** *Let $D \geq 1$, $m \neq 0$, $B \geq 1$ be integers and $Z = B(\lfloor\sqrt{D}\rfloor + 2)$. The number of $(x,y) \in [-B,B]^2$ with $x^2 - Dy^2 = m$ is at most $2\tau(|m|)^3(\log_3 Z^2 + 1)$.*

Proof. If $D = h^2$ then $(x-hy)(x+hy) = m$, and $x - hy$ is one of the $2\tau(|m|)$ signed divisors of m; it determines (x,y). Otherwise write $D = df^2$ with $d > 1$ squarefree, so d is not a square and $\sqrt{d}$ is irrational. For a solution put $g = \gcd(x, fy)$, $u = x/g$, $w = fy/g$ and $n = u^2 - dw^2$, so that $m = g^2n$. Since $\gcd(u,w) = 1$ and $n \equiv u^2 \pmod w$, w is invertible modulo n. Let $t \in [0, |n|)$ be the residue of uw^{-1}; then $t^2 \equiv d$ and $u \equiv tw$ modulo n.

The residues. The number of square roots of d modulo $c \geq 1$ is at most $\tau(c)^2$. Modulo p^e it is at most four. If $p \mid d$, either $e = 1$ and the only root is 0, or $e \geq 2$ and a root would force $p^2 \mid d$. If $p \nmid d$ is odd, two roots r, s satisfy $p^e \mid (s-r)(s+r)$, and p cannot divide both factors, so $s \equiv \pm r$. If $p = 2 \nmid d$, then s, r are odd, one of $(s \mp r)/2$ is odd, and $s \equiv \pm r \pmod{2^{e-1}}$. The count is submultiplicative over coprime factors, by the Chinese remainder theorem, and $4 \leq \tau(p^e)^2$.

A class. Fix g (with $g^2 \mid m$), t and the sign of $\xi = u + w\sqrt{d}$. Two solutions $(u, w), (u', w')$ in this class satisfy $n \mid uu' - dww'$ and $n \mid wu' - uw'$. So $\xi = \xi'\varepsilon$ with $\varepsilon = r + s\sqrt{d}$, where $r = (uu' - dww')/n$, $s = (wu' - uw')/n$ and $r^2 - ds^2 = 1$. If $|\xi| > |\xi'|$ then $\varepsilon > 1$, and then $\varepsilon \geq 3$: indeed $\varepsilon^{-1} = r - s\sqrt{d} \in (0, 1)$, so $2r = \varepsilon + \varepsilon^{-1} > 2$, hence $r \geq 2$, and $\varepsilon < 3$ would give $\varepsilon + \varepsilon^{-1} < 4$. Now $|u| + |w|\sqrt{d} \leq |x| + |y|\sqrt{D} \leq Z$, so $|\xi|, |\bar{\xi}| \leq Z$, and $|\xi\bar{\xi}| = |n| \geq 1$ gives $Z|\xi| \geq 1$. Hence the map $\xi \mapsto \lfloor \log_3(Z|\xi|) \rfloor$ is injective on the class, with values in $[0, \log_3 Z^2]$.

There are at most $\tau(|m|)$ values of g, $\tau(|m|)^2$ of t and two signs. □

The argument is the classical one; compare Vaughan and Wooley [14, Lemma 3.5].

12.4. **Slices.**

Lemma 12.4 (Slices)**.** *Let $F(x, y, z) = \alpha x^2 + \beta xy + \gamma y^2 + (d_1 z + d_0)x + (e_1 z + e_0)y + f_2 z^2 + f_1 z + f_0$ have integer coefficients bounded by K, with $\alpha > 0$, $D = \beta^2 - 4\alpha\gamma > 0$ and $\ell = (\beta d_1 - 2\alpha e_1)^2 + D(4\alpha f_2 - d_1^2) \neq 0$. Let $H \geq 1$, and suppose that $2\tau(|m|)^3(\log_3\big((11K^2H(3K+2))^2\big) + 1) \leq \omega$ for all integers $0 < |m| \leq 116K^4H^2$. Then F has at most $(2H+1)(\omega+4)$ zeros in $[-H, H]^3$.*

Proof. Put $t = 2\alpha x + \beta y + d(z)$, $\Lambda(z) = \beta d(z) - 2\alpha e(z)$, $s = Dy + \Lambda(z)$ and $\mathcal{N}(z) = \Lambda(z)^2 + D(4\alpha f(z) - d(z)^2)$. Completing squares gives $s^2 - Dt^2 = \mathcal{N}(z) - 4\alpha DF$. In the box, $|s|, |t| \leq 11K^2H$, $\sqrt{D} \leq 3K$ and $|\mathcal{N}(z)| \leq 116K^4H^2$. For fixed z the map $(x, y) \mapsto (s, t)$ is injective. So if $\mathcal{N}(z) \neq 0$ the slice has at most ω zeros, by Lemma 12.3. The z^2-coefficient of $\mathcal{N}$ is $\ell \neq 0$, so at most two slices have $\mathcal{N}(z) = 0$. In such a slice $s^2 = Dt^2$, which for fixed y is a quadratic in x with leading coefficient $4\alpha^2 D \neq 0$, so it has at most $2(2H+1)$ zeros. □

12.5. **The count.**

Proposition 12.5. *Under the hypotheses of Proposition 11.7, let $K = 4M^2$, and suppose ω satisfies the hypothesis of Lemma 12.4. Then $|\mathcal{G}| \leq 4(2M+1)^7(2H+1)(\omega+4)$.*

Proof. As before, at most four states share (θ, κ), and κ is determined by (θ, k_1, k_2, k_3). For a fixed θ, the triples (k_1, k_2, k_3) are zeros of F_θ in $[1, H]^3$ (Lemma 12.2). The coefficients of F_θ are at most $4M^2$, and Lemmas 12.1 and 12.2 give the nondegeneracy hypotheses of Lemma 12.4. □

Let $P_1(a) = 10^{300}a^{100}$, let $\tau^*(X) = \max_{n \leq X} \tau(n)$ with the convention $\tau(0) = 0$, and let $\Xi(a) = 2\tau^*(P_1(a))^3(\log_3 P_1(a) + 1) + 4$. In every window $H \leq 29a$ and $M \leq 10^{16}a^7$; the same bound holds for the larger M of Proposition 13.4 below. Hence $K = 4M^2$ satisfies $116K^4H^2 \leq P_1(a)$ and $(11K^2H(3K+2))^2 \leq P_1(a)$, so $\omega = \Xi(a) - 4$ satisfies the hypothesis of Lemma 12.4. The argument of Proposition 11.8 then gives:

Proposition 12.6. *In the setting of Proposition 11.8, if* (11) *holds then $T \leq 24(2M+1)^7 H\, \Xi(a)$.*

Proposition 12.7. *For every window run with $T \geq 36$, $T^{74} \leq C_{74}\Delta^{19}\Xi(a)^{19}a^{22}$ for an absolute constant C_{74}.*

Proof. We prove this with $C_{74} = 122449^{19}+(8\cdot237568\cdot56^5)^{11}+10752^{19}+8^{15}\cdot24\cdot3^7\cdot172032^7\cdot56^{43}$. The four cases of Proposition 11.9 carry over. If $AT > 28a$, then $T^4 \le 122449a$. If the scale condition fails, then $T^7 \le 8\cdot237568\cdot56^5\Delta a^2$. If $M = 0$, then $T^2A \le 1344\,\Xi a$ and $T^4 \le 10752\,\Delta\Xi a$. If $M \ge 1$, then $Ta^{21} \le 24\cdot3^7\cdot172032^7A^{28}H^{43}\Xi$, so $T^{44}A^{15} \le 24\cdot3^7\cdot172032^7\cdot56^{43}\,\Xi a^{22}$, and $T^{74} = T^{44}(T^2)^{15} \le (8A\Delta)^{15}T^{44}$. □

Proof of Theorem 1.6. By the divisor bound $\tau(n)^k \le C_k n$, both Δ and $\tau^*(P_1(a))$ are $O_\eta(a^\eta)$, and $\log_3 P_1(a) \ll_\eta a^\eta$. So $\Xi(a) \ll_\eta a^\eta$, and Proposition 12.7 gives $T \ll_\eta a^{(22+38\eta)/74}$ for every dyadic window. Since $22/74 = 11/37$, summing over the $O(\log a)$ windows proves the theorem. □

Remark 12.8. In terms of the scales, the count per pattern is now $O(Ha^\eta)$ instead of $O(H^2)$. So $T \ll a^\eta X^7 H$, which gives $T^{44} \ll a^{7+\eta}B^{15}$ with $B = a/A$, and with $B \ll a^{1+\eta}/T^2$ the bound $T^{74} \ll a^{22+O(\eta)}$. At the critical point $T \approx a^{11/37}$, $B \approx a^{15/37}$, $H \approx a^{4/37}$ and $X \approx a^{1/37}$.

13. Ten minors and the exponent 8/27

The seven pattern integers of Section 12 were counted as if they were independent. They are not: they are coordinates of a single vector of 3×3 minors. Counting such vectors directly saves a factor X^2 and proves Theorem 1.7.

13.1. **The full minor vector.** Let $\mathcal{M}_i$ now be the 3×5 matrix with rows $(d_0,\dots,d_4)$, $(k_0,\dots,k_4)$ and $(r_0,\dots,r_4)$, and let $p_{c_1c_2c_3}$ be its minor on columns $c_1<c_2<c_3$. Expanding the cofactors of Section 11 gives

$$\theta(i) = (p_{012},\, p_{123},\, p_{234},\, p_{023},\, -p_{013},\, p_{134},\, -p_{124}).$$

Lemma 13.1 (Expansion)**.** *Let $q = q_i$ and $s_j = q_{i+j} - q$. If $i+j+3 < N$ then*

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{1+s_j}{q^2} + \frac{s_j^2+2s_j+k_j+1}{q^3} - e_j, \qquad 0 \le e_j \le \frac{\mu_{i+j}\big((x+s_j)^3\big)}{q^4}.$$

Proof. Apply μ_{i+j} to $\frac{1}{q+y} = \frac{1}{q} - \frac{y}{q^2} + \frac{y^2}{q^3} - \frac{y^3}{q^3(q+y)}$ with $y = s_j + x \ge 0$, using $b_{i+j}/a = \nu_{i+j}(1) = \mu_{i+j}\big(1/(q+s_j+x)\big)$, $\mu(1) = \mu(x) = 1$ and $\mu(x^2) = k+1$. □

Proposition 13.2 (All minors)**.** *Let $i+8<N$, $b_i < 2A$ and $k_0,\dots,k_6 \le H$ at state i. Then every minor of $\mathcal{M}_i$ satisfies $|p|\,a^3 \le 3609600\,A^4H^6$.*

Proof. Subtracting the expansions for j and $j+1$ gives $d_j/a = k_j/q^2 - (r_j+k_j)/q^3 + e_{j+1} - e_j$. So replacing the first row by $d'_j = d_j - a(q^{-2}-q^{-3})k_j + aq^{-3}r_j = a(e_{j+1}-e_j)$ changes no minor. Now $|d'_j| \le 235aH^3/q^4$, $|k_j| \le H$ and $|r_j| \le 10H^2$, so $|p| \le 14100\,aH^6/q^4$. Finally $a \le 4Aq$. □

This replaces Lemma 11.6, and the scale condition (11) is no longer needed.

13.2. **Counting minor vectors.**

Proposition 13.3 (Minor vectors)**.** *The number of vectors of minors of integer 3×5 matrices with all coordinates in $[-Y,Y]$ is at most $1 + 20\sum_{m\le Y} 729\,m^4\tau(m) \ll Y^{5+o(1)}$.*

Proof. For a nonzero vector p choose a pivot π with $|p_\pi| = |m|$ maximal, and let c_4, c_5 be the two columns outside π. Let u_a (v_a) be the minor with the a-th pivot column replaced by c_4 (c_5), with the sign of the column permutation. By Cramer's rule and the Plücker relations, the remaining three minors satisfy $m\,p_{w_a} = \pm(u\times v)_a$ (thirty polynomial identities, checked

symbolically). So p is determined by $(\pi, \operatorname{sgn} m, |m|, u, v)$, where $|u_a|, |v_a| \le |m|$ and $|m|$ divides every coordinate of $u \times v$.

Fix $m \ge 1$ and u, and let $g = \gcd(m, u_1, u_2, u_3)$, $n = m/g$, $u = gu_0$. By Bézout there are $c \in \mathbb{Z}^3$ and $x \in \mathbb{Z}$ with $c \cdot u_0 + xn = 1$. If $m \mid u \times v$, then $n \mid u_0 \times v$, and the identity $v - (c \cdot v)u_0 = xn\, v + c \times (v \times u_0)$ shows $v \equiv (c \cdot v)u_0 \pmod n$. Hence v lies in one of n classes modulo n, each meeting $[-m, m]^3$ in at most $(3g)^3$ points: at most $27mg^2$ choices. Summing over u, grouped by $g = d \mid m$, gives $\sum_u g^2 \le \sum_{d|m} d^2(3m/d)^3 \le 27m^3\tau(m)$, so at most $729m^4\tau(m)$ pairs (u, v). There are ten pivots and two signs. □

13.3. **The bound.**

Proposition 13.4. *For a window run in $[A, 2A)$ with $T \ge 36$, let $H = \lfloor 28a/(AT) \rfloor + 1$ and $M = \lfloor 3609600A^4H^6/a^3 \rfloor$. Then $T \ll (M+1)^5 H\, \Xi(a)^2$ and*

$$T^{54} \ll \Delta^{14}\Xi(a)^{28}a^{16}.$$

Proof. Explicitly, we prove $T \le 24 \cdot 14581\,(M+1)^5 H\, \Xi(a)^2$ and $T^{54} \le C_{54}\Delta^{14}\Xi(a)^{28}a^{16}$ with $C_{54} = 122449^{14} + (448 \cdot 24 \cdot 14581)^{14} + 8^{11} \cdot 32 \cdot 24 \cdot 14581 \cdot 3609600^5 \cdot 56^{31}$. As in Propositions 11.8 and 12.5, at least $T/2$ starts are good, at most four good starts share (θ, κ), and there are at most $(2H+1)(\omega+4)$ gap triples per pattern. Since θ is a function of the minor vector, there are at most $1 + 14580M^5\tau^*(M) \le 14581(M+1)^5\Xi(a)$ patterns, by Propositions 13.2 and 13.3. For the second bound, if $AT > 28a$ then $T^4 \le 122449a$. Otherwise $HAT \le 56a$. If $M = 0$ then $T^4 \le 448 \cdot 24 \cdot 14581\,\Delta\Xi^2 a$. Otherwise $Ta^{15} \le 32 \cdot 24 \cdot 14581 \cdot 3609600^5 A^{20}H^{31}\Xi^2$, hence $T^{32}A^{11} \ll \Xi^2 a^{16}$ and $T^{54} \le (8A\Delta)^{11}T^{32}$. □

Proof of Theorem 1.7. Since $\Delta, \Xi(a) \ll_\eta a^\eta$, every window has $T \ll a^{(16+42\eta)/54}$, and $16/54 = 8/27$. Choose η sufficiently small in terms of ε, include the bounded-length windows, and sum over the $O(\log a)$ dyadic windows. □

Remark 13.5. In terms of the scales, $T \ll a^\eta X^5 H$, so $T^{32} \ll a^{5+\eta}B^{11}$ and $T^{54} \ll a^{16+O(\eta)}$. At the critical point $T \approx a^{8/27}$, $B \approx a^{11/27}$, $H \approx a^{3/27}$ and $X \approx a^{1/27}$.

14. Lattices of minor vectors and the exponent 45/152

Proposition 13.4 allows $(2H+1)(\omega+4)$ gap vectors for every minor vector. But the gap vector of a state is not an arbitrary point of $[1, H]^5$: it lies in a rank-three lattice determined by its minor vector. If that lattice has no short vectors, the slices of the ternary quadratic are few. If it has a short vector, the minor vector satisfies an extra congruence, and few minor vectors do. Balancing the two proves Theorem 1.8. The argument uses only integer arithmetic. Instead of the geometry of numbers it uses a reduced basis obtained by minimising determinants and lengths directly.

14.1. **The lattice of a minor vector.** Let $p \ne 0$ be the minor vector of a 3×5 integer matrix $\mathcal{M}$, with pivot π, $m = p_\pi$, pivot columns $c_1 < c_2 < c_3$, other columns $c_4 < c_5$, and u, v as in Proposition 13.3. Let $\Lambda_p \subseteq \mathbb{Z}^3$ be the subgroup of x with $m \mid u \cdot x$ and $m \mid v \cdot x$. It contains $m\mathbb{Z}^3$. Let $\lambda(p)$ be the least sup-norm of a nonzero vector of Λ_p. Subscripts on L_c and p_{abc} retain the original column labels $0, \dots, 4$; c_1, c_2, c_3 enumerate the chosen pivot. We use $\gcd(n, x)$ for $\gcd(n, x_1, x_2, x_3)$ when $x \in \mathbb{Z}^3$.

Lemma 14.1 (Lifting)**.** *Let L_c be me_a if $c = c_a$ is a pivot column, u if $c = c_4$ and v if $c = c_5$. For every row x of $\mathcal{M}$ and every column c, $m\, x_c = L_c \cdot (x_{c_1}, x_{c_2}, x_{c_3})$. Moreover $\det(L_1, L_2, L_3) = m^2 p_{123}$.*

Proof. The rows L_c are the rows of $\mathcal{M}^{\mathsf{T}}\operatorname{adj}(\mathcal{M}_\pi^{\mathsf{T}})$, where $\mathcal{M}_\pi$ is the pivot 3×3 block. This is Cramer's rule. Both claims are polynomial identities in the fifteen entries, checked symbolically for each of the ten pivots. □

So the pivot gaps $k' = (k_{c_1}, k_{c_2}, k_{c_3})$ of a state lie in Λ_p. The five gaps of the state are $k_c = L_c \cdot k'/m$, and since $p_{123} = L(i+1) \geq 1$, the gaps (k_1, k_2, k_3) are an invertible linear image of k'.

Lemma 14.2 (Index)**.** *If t is a nonnegative integer and $(t+1)^3 > |m|$, then $\lambda(p) \leq t$.*

Proof. Write $g = \gcd(|m|, u)$, $n = |m|/g$ and $u = gu_0$. The proof of Proposition 13.3 gives $v = \mu u_0 + nw$ with $\mu = c\cdot v \in \mathbb{Z}$ and $w \in \mathbb{Z}^3$. Put $\psi(x) = \big(u_0\cdot x \bmod n,\ (\mu\lfloor u_0\cdot x/n\rfloor + w\cdot x) \bmod g\big)$, which takes at most $ng = |m|$ values. Two points $x \neq y$ of $[0,t]^3$ with $\psi(x) = \psi(y)$ exist. Then $u_0\cdot(x-y) = n(q-q')$ with q, q' the two floors, so $u\cdot(x-y) = \pm m(q-q')$, and $v\cdot(x-y) = n\big(\mu(q-q') + w\cdot(x-y)\big)$ with the bracket divisible by g. So $x - y \in \Lambda_p$. □

Since $\psi(x) = \psi(y)$ forces $x - y \in \Lambda_p$, the same argument shows that Λ_p has index at most $|m|$ in $\mathbb{Z}^3$.

14.2. **Reduced bases.** Here $|v|$ denotes the Euclidean norm, while $|v|_\infty$ denotes the sup norm used to define $\lambda(p)$. In particular, $|v| \geq |v|_\infty$.

Lemma 14.3 (Reduction)**.** *Let $\Lambda \subseteq \mathbb{Z}^3$ be a subgroup containing $m\mathbb{Z}^3$, $m \geq 1$. There is a basis b_1, b_2, b_3 of Λ with $\delta = \det(b_1, b_2, b_3)$, $0 < |\delta| \leq m^3$ and*

$$27\,|b_1|^2|b_2|^2|b_3|^2 \leq 64\,\delta^2.$$

Hence every $w \in \Lambda$ is $\sum y_i b_i$ with $y \in \mathbb{Z}^3$ and $27|b_i|^2 y_i^2 \leq 64|w|^2$.

Proof. Let I be the least nonzero $|\det|$ of a triple in Λ; the triple (me_1, me_2, me_3) gives $I \leq m^3$. A triple with $|\det| = I$ is a basis. Otherwise some $w \in \Lambda$ has a Cramer coordinate that is not an integer, say the first, $\det(w, b_2, b_3)/\delta$, and $w - \lfloor\cdot\rfloor b_1$ gives a triple with smaller nonzero determinant. Among such bases minimise $d_1 = |b_1|^2$, and then the Gram determinant $d_2 = d_1|b_2|^2 - \langle b_1, b_2\rangle^2$. Put $d_3 = \delta^2$.

Choose n with $|2(\langle b_1, b_2\rangle + nd_1)| \leq d_1$. Then $(b_2 + nb_1, b_1, b_3)$ is a basis, so $d_1 \leq |b_2 + nb_1|^2$, and $d_1|b_2 + nb_1|^2 = d_2 + (\langle b_1, b_2\rangle + nd_1)^2$. Hence $3d_1^2 \leq 4d_2$.

Let $B(y,z) = d_1\langle y, z\rangle - \langle b_1, y\rangle\langle b_1, z\rangle$, so that $B(b_2, b_2) = d_2$, and Sylvester's identity gives $B(b_2,b_2)B(b_3,b_3) - B(b_2,b_3)^2 = d_1 d_3$. Choose n_2 with $|2(B(b_2,b_3) + n_2 d_2)| \leq d_2$ and put $c = b_3 + n_2 b_2$. Then (b_1, c, b_2) is a basis with first vector b_1, so $d_2 \leq d_1|c|^2 - \langle b_1, c\rangle^2 = B(c,c)$, while $d_2 B(c,c) = (B(b_2,b_3) + n_2 d_2)^2 + d_1 d_3$. Hence $3d_2^2 \leq 4d_1 d_3$.

Finally choose n_1 with $|2(\langle b_1, c\rangle + n_1 d_1)| \leq d_1$, and use the basis b_1, $b_2' = b_2 + nb_1$, $b_3' = c + n_1 b_1$. The identities above give $3d_1|b_2'|^2 \leq 4d_2$ and $4d_1 d_2|b_3'|^2 \leq d_2^2 + 4d_1 d_3 + d_1^2 d_2$. With the two inequalities this gives $9d_2|b_3'|^2 \leq 16 d_3$, and the product of the three bounds is the claim. For the coordinates, $y_1\delta = \det(w, b_2, b_3)$ and Hadamard's inequality $\det(w, b_2, b_3)^2 \leq |w|^2|b_2|^2|b_3|^2$. □

14.3. **Few minor vectors have a short vector.**

Lemma 14.4. *For $n \geq 1$ and $x \in \mathbb{Z}^3$, at most $27n^2\gcd(n, x)$ vectors $w \in [-n, n]^3$ have $n \mid w\cdot x$.*

Proof. Reducing modulo n loses a factor at most 27. Let K be the set of solutions in $[0, n)^3$, $e = \gcd(n, x)$ and $c\cdot x \equiv e \pmod n$ by Bézout. The map $(w, j) \mapsto (w + jc) \bmod n$ is injective on $K \times [0, n/e)$: equal images give $n \mid (j - j')e$, so $j = j'$, and then $w = w'$. Hence $|K|\, n/e \leq n^3$. □

Proposition 14.5 (Short vectors)**.** *Let $\mathcal{S}$ be a set of nonzero minor vectors with coordinates in $[-Y, Y]$, and let $\tau(m) \le \tau_0$ for $m \le Y$. Then for every $R \ge 1$,*

$$\#\{p \in \mathcal{S} : \lambda(p) \le R\} \le 393660\, R^3 Y^4 \tau_0^2.$$

Proof. If $\lambda(p) \le R$, some nonzero $x \in [-R, R]^3$ has $m \mid u \cdot x$. For fixed x, the encoding of Proposition 13.3 gives at most $20 \sum_m N(m, x)$ such p, where $N(m, x)$ counts pairs (u, v) with $m \mid u \times v$ and $m \mid u \cdot x$. For each u there are at most $27m \gcd(m, u)^2$ vectors v. Grouping the u by $g = \gcd(m, u)$ and writing $u = gu'$ with $(m/g) \mid u' \cdot x$, Lemma 14.4 gives $\sum_u g^2 \le 27m^2\tau(m) \gcd(m, x)$, so $N(m, x) \le 729m^3\tau(m) \gcd(m, x)$. Finally $\sum_{0<|x|_\infty \le R} \gcd(m, x) \le 27R^3\tau(m)$. A divisor $h > R$ of m divides no nonzero x in the box, and for $h \le R$ the multiples of h number at most $(2R/h + 1)^3$, with $h(2R/h + 1) \le 3R$. □

14.4. **One minor vector.**

Lemma 14.6 (Definite norm equations)**.** *For $D \ge 1$ and $m \ne 0$, the equation $x^2 + Dy^2 = m$ has at most $4\tau(|m|)^3$ integer solutions.*

Proof. As in Lemma 12.3, with $D = df^2$, d squarefree, each solution reduces to a primitive $u^2 + dw^2 = n$, $m = g^2 n$, with a class $t \equiv uw^{-1}$, a square root of $-d$ modulo n. The count of square roots in the proof of Lemma 12.3 applies to $-d$ unchanged. Two solutions in a class give integers r, s with $r^2 + ds^2 = 1$, of which there are at most four. □

A ternary quadratic F with quadratic part of Gram matrix S has $\det(2S)$ multiplied by $\det(C)^2$ under a substitution $x = Cy$. In the notation of Lemma 12.4 one has the identity $\ell = -2\alpha \det(2S)$. For F_θ, Lemma 12.2 therefore gives $\det(2S) = -8N_0N_1^2(N_0 - u_2)(N_1 + u_1)(u_1 - u_2) \ne 0$ under the hypotheses of Lemma 12.1, so the following lemma applies to F_θ after any invertible linear change of variables.

Lemma 14.7 (Plane substitution)**.** *Let F be a ternary quadratic with integer coefficients whose quadratic part has Gram matrix S with $\det(2S) \ne 0$. There is a substitution $x = Cy$, where C is a product of two integer matrices with entries in $\{0, \pm 1\}$, determinant ± 1 and inverses with absolute row sums at most 2, such that $F(Cy)$ has nonzero y_1^2 coefficient α, nonzero binary discriminant D in (y_1, y_2), and $\ell \ne 0$, in the notation of Lemma 12.4. Moreover, the conclusion of Lemma 12.4 holds with $D > 0$ replaced by $D \ne 0$, under the same hypothesis on ω.*

Proof. Since $\ell = -2\alpha \det(2S)$ and $\det(2S)$ changes only by the factor $\det(C)^2 = 1$, it suffices to make α and D nonzero. First choose one of the planes $\langle e_1, e_2\rangle$, $\langle e_1, e_3\rangle$, $\langle e_2, e_3\rangle$, $\langle e_1 + e_2, e_3\rangle$: if all four binary discriminants vanished then $\det(2S)$ would vanish, by an explicit combination with polynomial coefficients. Writing f, g for the displayed spanning vectors of the chosen plane, choose f, g or $f + g$ as its first basis vector. With this, Lemma 12.4 holds for $\alpha \ne 0$ and $D \ne 0$ of either sign, under the same hypothesis on ω. When $D < 0$ each slice uses Lemma 14.6, whose bound $4\tau(|m|)^3$ is at most $2\tau(|m|)^3(\log_3\big((11K^2H(3K+2))^2\big) + 1)$ because $11K^2H(3K + 2) \ge 3$. □

Proposition 14.8 (One minor vector)**.** *Let p be a minor vector with all coordinates in $[-X, X]$, $X \ge 1$, and suppose ω satisfies the hypothesis of Lemma 12.4 for $K = 248832X^8$ and $12H + 1$ in place of H. The good starts with minor vector p have at most*

$$\big(8\lfloor 3H/\lambda(p)\rfloor + 3\big)(\omega + 4)$$

gap vectors κ.

Proof. At a good start the gaps lie in $[1, H]$, and since $N_0 = L(i)$ and $N_1 = L(i+1)$ are positive integers, Lemma 12.1 gives $u_1 > 0 > u_2$. Let b_1, b_2, b_3 be the basis of Lemma 14.3 for Λ_p, with $|m|$ in place of m, and $\hat{b}_i$ their lifts $c \mapsto L_c \cdot b_i/m$ to $\mathbb{Z}^5$. Write $k' = \sum y_i b_i$. By Lemma 14.1, $\kappa = \sum y_i \hat{b}_i$, so $(k_1, k_2, k_3) = \hat{B}y$ for the 3×3 matrix $\hat{B}$ of the lifts in columns $1, 2, 3$, and $m \det \hat{B} = p_{123}\delta \neq 0$. So y is a zero of $\tilde{F} = F_\theta \circ \hat{B}$, and $\det(2S_{\tilde{F}}) = \det(2S_{F_\theta})(\det \hat{B})^2 \neq 0$. Each $|b_i|^2 \geq \lambda(p)^2$ and $|k'|^2 \leq 3H^2$, so $\lambda(p)|y_i| \leq 3H$, and after the plane substitution the coordinates lie in $[-Y', Y']$ with $Y' = 4\lfloor 3H/\lambda(p) \rfloor + 1 \leq 12H + 1$ (Lemma 14.7). Every nonzero integer basis vector has length at least one, so Lemma 14.3 gives $|b_i| \leq 2X^3$. Maximality of the pivot bounds each entry of L_c by $|m|$. Thus every lift is at most $6X^3$, and the coefficients of the final quadratic are at most $12^3 \cdot 4X^2 \cdot (6X^3)^2 = 248832X^8$. Lemma 12.4, in the form of Lemma 14.7, therefore bounds the zeros by $(2Y' + 1)(\omega + 4)$, and y determines κ. □

14.5. **The bound.**

Lemma 14.9 (Partial summation)**.** *Let t and R_p for $p \in \mathcal{S}$ be integers with $1 \leq R_p \leq t$, and let $\#\{p : R_p \leq r\} \leq Cr^3$ for $r \geq 1$. Then $t \sum_p \lfloor n/R_p \rfloor \leq n|\mathcal{S}| + nCt^3$ for every $n \geq 0$.*

Proof. Write $1/r = 1/t + \sum_{r \leq j < t} 1/(j(j+1))$ and exchange the sums: $\sum_p 1/R_p \leq |\mathcal{S}|/t + \sum_{j<t} Cj^3/(j(j+1)) \leq |\mathcal{S}|/t + Ct^2$. □

Let $P_2(a) = 10^{1000}a^{400}$ and $\Xi_2(a) = 2\tau^*(P_2(a))^3(\log_3 P_2(a) + 1) + 4$. As in Section 12, $\omega = \Xi_2(a) - 4$ is admissible in every window, and $\Xi_2(a) \ll_\eta a^\eta$.

Proposition 14.10. *For a window run in $[A, 2A)$ with $T \geq 36$, let $H = \lfloor 28a/(AT) \rfloor + 1$ and $X = \lfloor 3609600A^4H^6/a^3 \rfloor + 1$. Then*

$$T^3 \ll \Xi_2(a)^9 \, (H^3X^{14} + X^{15}) \qquad \text{and} \qquad T^{152} \ll \Delta^{39}\Xi_2(a)^{117}a^{45}.$$

Proof. Explicitly, we prove $T^3 \leq C_\star \, \Xi_2(a)^9 \, (H^3X^{14} + X^{15})$ with $C_\star = 32 \cdot 607461312^3$, and $T^{152} \leq C_{152}\Delta^{39}\Xi_2(a)^{117}a^{45}$ with $C_{152} = 122449^{38} + c_0^{13} + 8^{31}c_A + 8^{30}c_B$, where $c_0 = 1024\, C_\star \, 56^3$, $c_A = 2^{15}C_\star \, 3609600^{14}56^{87}$ and $c_B = 2^{16}C_\star \, 3609600^{15}56^{90}$. Let $\mathcal{S}$ be the set of minor vectors of good starts and $\tau_0 = \tau^*(P_2(a))$. At most four good starts share (p, κ), and $T \leq 2|\mathcal{G}|$, so Proposition 14.8 gives

$$T \leq 8\,\Xi_2(a)\Big(8\sum_{p \in \mathcal{S}} \lfloor 3H/\lambda(p) \rfloor + 3|\mathcal{S}|\Big).$$

Choose t with $X \leq t^3 \leq 8X$. Lemma 14.2 gives $\lambda(p) \leq t$, and Proposition 14.5 and Lemma 14.9 give $t \sum_p \lfloor 3H/\lambda(p) \rfloor \leq 3H|\mathcal{S}| + 3H \cdot 393660X^4\tau_0^2 t^3$, with $|\mathcal{S}| \leq 14581X^5\tau_0$. Hence $Tt \leq 607461312\,\Xi_2^3X^5(H + t)$, and cubing, with $(H + t)^3 \leq 4(H^3 + t^3)$ and $X \leq t^3 \leq 8X$, gives the first bound.

For the second, if $AT > 28a$ then $T^4 \leq 122449a$. Otherwise $H\,AT \leq 56a$. If $X = 1$ then $T^6A^3 \leq 2C_\star \, 56^3\Xi_2^9a^3$, and $T^{12} \leq (8A\Delta)^3T^6 \leq c_0\Delta^3\Xi_2^9a^3$. If $X \geq 2$, write $X = X_0 + 1 \leq 2X_0$ with $X_0a^3 \leq 3609600A^4H^6$. When $2X_0 \leq H^3$, $T^3a^{42} \ll \Xi_2^9A^{56}H^{87}$, so $T^{90}A^{31} \leq c_A\Xi_2^9a^{45}$ and $T^{152} \leq (8A\Delta)^{31}T^{90}$. When $2X_0 > H^3$, $T^3a^{45} \ll \Xi_2^9A^{60}H^{90}$, so $T^{93}A^{30} \leq c_B\Xi_2^9a^{45}$ and $T^{153} \leq (8A\Delta)^{30}T^{93}$. □

Proof of Theorem 1.8. Since $\Delta, \Xi_2(a) \ll_\eta a^\eta$, every window has $T \ll a^{(45+156\eta)/152}$. Summing over the $O(\log a)$ windows proves the theorem. □

Remark 14.11. In terms of the scales, $T \ll a^\eta(HX^{14/3} + X^5)$: the factor $X^{1/3}$ saved over Proposition 13.4 is the typical $\lambda(p)$. At the critical point $T \approx a^{45/152}$, $B \approx a^{62/152}$, $H \approx a^{17/152}$ and $X \approx a^{6/152}$, and the term $X^5 \approx a^{30/152}$ is smaller.

15. Joint integer invariants and the optimized bound

We now prove Theorem 1.1. Fix a window run in $[A, 2A)$ with $T \geq 36$ terms. Throughout this section constants may depend on fixed positive integers m, n (the powers of Section 15.9) and on a loss $\eta > 0$, but never on the dividend, the orbit or the pattern. The argument has six steps.

(1) For a fixed pattern, the nodes of every good start lie on at most 24 curves (Section 15.1).
(2) On these curves the quotient q_i and the curvature $I_3(i)$ are rational functions of the nodes. Components on which the reconstruction identities fail contribute $O(1)$ starts (Section 15.2).
(3) The directions at infinity of the curves are controlled by the Newton coefficients of the pattern. This makes nonexceptional components have even degree (Section 15.3), shows that exceptional patterns are few, and excludes lines (Section 15.4).
(4) The reconstructed functions, and hence two families of new integer coordinates W and J, have prescribed orders at infinity (Sections 15.5 and 15.6).
(5) Each nonexceptional conic contributes $a^{o(1)}$ starts (Section 15.7).
(6) On the remaining components, adjoining W_j^m and J^n raises the degree. The curve-counting theorem and a balancing of heights then give the exponent (Sections 15.8 and 15.9).

Notation. The symbols D, L and W of earlier sections do not appear here with their earlier meanings. In this section

- $\theta = (N_0, N_1, N_2, u_1, u_2, v_1, v_2)$ are the pattern integers of Lemma 11.3, with $N_0 = I_4(i)$, and bars denote their normalised versions (12);
- s_j are the nodes, k_j the gaps, P_4, R_4 the two orthogonal quartics, Ψ the quartic norm and f the pattern quadratic;
- Q and E_0 are the reconstructed quotient and curvature, $W = (W_0, W_1, W_2)$ and J the joint coordinates, and S, Y their height bounds;
- e is a multiplicity at infinity, d the degree of a component, δ the degree of a graph and $\delta_{\min} = 10n + 7m$.

15.1. Good starts and the pattern curve.

Put

$$H = \left\lfloor \frac{28a}{AT} \right\rfloor + 1, \qquad X = \left\lfloor \frac{3609600A^4H^6}{a^3} \right\rfloor.$$

As in Proposition 11.8, call a start i *good* if the window contains the nine states $i, \dots, i+8$ and the gaps $k_0, \dots, k_6$ at state i are at most H; at least $T/2$ starts are good. By Proposition 13.2, the minor vector of a good start has all coordinates at most X, and $X \geq 1$ because its coordinate $p_{012} = L(i)$ is a positive integer. We call this minor vector the *pattern* of the start; it determines $\theta(i)$. By Proposition 13.3, there are $O_\eta(a^\eta X^5)$ patterns.

The *nodes* of state i are $s_j = q_{i+j} - q_i$. They are the support points of the measure μ_i of Section 10, and the interpolation nodes of the Newton polynomials of Section 11. The three *free nodes* s_1, s_2, s_3 of a good start are integers in $[0, 3H]$. Distinct good starts with the same pattern have distinct free-node triples (Lemma 15.2 below).

For the pattern θ, put

$$(\bar{N}_1, \bar{u}_1, \bar{u}_2) = \frac{(N_1, u_1, u_2)}{N_0}, \qquad (\bar{N}_2, \bar{v}_1, \bar{v}_2) = \frac{(N_2, v_1, v_2)}{N_1}. \tag{12}$$

By Lemma 12.1, $\bar{N}_1 > 0$, $\bar{u}_1 > 0$ and $\bar{u}_2 < 0$. Treat the free nodes s_1, s_2, s_3 as coordinates and define

$$s_0 = 0, \quad s_4 = (\bar{N}_1 + \bar{u}_1)s_1 - (\bar{u}_1 - \bar{u}_2)s_2 - (\bar{u}_2 - 1)s_3, \quad k_j = s_{j+1} - s_j \quad (0 \le j \le 3),$$
$$k_4 = \bar{N}_2 k_1 - \bar{v}_1 k_2 - \bar{v}_2 k_3, \qquad s_5 = s_4 + k_4.$$

At a good start these are the actual nodes and gaps, by the two k-relations of Lemma 11.1; the first is equivalent to $\bar{N}_1 k_0 = \bar{u}_1 k_1 + \bar{u}_2 k_2 + k_3$. Let

$$\gamma = (\bar{N}_1,\ -\bar{N}_1 - \bar{u}_1,\ \bar{u}_1 - \bar{u}_2,\ \bar{u}_2 - 1,\ 1),$$

and let γ' be the same expression in $\bar{N}_2, \bar{v}_1, \bar{v}_2$. Put

$$P_4(x) = \sum_{j=0}^{4} \gamma_j \prod_{\ell=0}^{j-1} (x - s_\ell), \qquad R_4(x) = \sum_{j=0}^{4} \gamma'_j \prod_{\ell=1}^{j} (x - s_\ell).$$

By the remark after Lemma 11.1, at a good start P_4 is the monic orthogonal polynomial π_i of degree four for ν_i, and $R_4(x) = \pi_{i+1}(x - k_0)$ is the corresponding polynomial of the next state, written in the variable of the current state. Put $Z = xR_4$ and $f = \sum_{j=0}^{4} \gamma_j (s_j^2 + k_j)$, the *pattern quadratic*, and let

$$\Psi = \sum_{j=0}^{4} \gamma_j \big(s_j^3 + 3s_j^2 + 3s_j(1 + k_j) + (1 + k_j)^2\big) + \sum_{j=0}^{3} \gamma_j k_{j+1}$$
$$+ \bar{N}_2 r_1 - \bar{v}_1 r_2 - \bar{v}_2 r_3 - k_4(2s_4 + k_4),$$

where $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$ for $j = 1, 2, 3$. This is a cubic polynomial in the free nodes, the *quartic norm*.

Lemma 15.1 (The pattern curve). (1) *At a good start i, $\Psi = \nu_i(P_4^2) > 0$, and the free nodes satisfy*

$$f = 0, \qquad N_0 \operatorname{Res}(P_4, Z) = a\Psi^4. \tag{13}$$

(2) *For every pattern with $\bar{N}_1, \bar{u}_1 > 0 > \bar{u}_2$, the two equations* (13) *have degrees two and at most twelve and no common factor. Their common zero set in $\mathbb{A}^3$ is the union of at most twenty-four integral affine curves over $\mathbb{Q}$, of total degree at most twenty-four.*

Proof. (1) Since $r_m = s_{m+1}^2 - s_m^2 + k_{m+1}$, summation by parts and the k-relation at state i show that $N_0 f$ is the r-relation $\sum_m U_m(i) r_m = 0$ of Lemma 11.1. Next, the shift rule $\mu_i\big(g \prod_{\ell<j}(x - s_\ell)\big) = \mu_{i+j}\big(g(x + s_j)\big)$ and the moments listed in the proof of Lemma 11.5 give $\mu_i(x^3 P_4) = \sum_j \gamma_j \big(s_j^3 + 3s_j^2 + 3s_j(1 + k_j) + (1 + k_j)^2 + k_{j+1}\big)$. Eliminating k_5 with the r-relation at state $i + 1$ turns this into Ψ. Since $\nu_i(xg) = \mu_i(g) - q_i \nu_i(g)$ and P_4 is orthogonal to x^3, $\nu_i(P_4^2) = \nu_i(x^4 P_4) = \mu_i(x^3 P_4) = \Psi$, and this is positive as in the proof of Theorem 10.2.

For the resultant, Lemma 10.1(3) gives $\nu_i\big(xg(x)\big) = \nu_{i+1}\big(g(x + k_0)\big)$. Hence, for $j < 4$, $\nu_i(x^j Z) = \nu_{i+1}\big((x + k_0)^j \pi_{i+1}(x)\big) = 0$. So the monic quintic Z differs from the monic orthogonal polynomial $\pi^{(5)}$ of degree five for ν_i by a multiple of P_4, and $\operatorname{Res}(P_4, Z) = \operatorname{Res}(P_4, \pi^{(5)})$. Let n_j be the squared ν_i-norm of the monic orthogonal polynomial of degree j. The three-term recurrence, whose coefficients are $\beta_j = n_j / n_{j-1}$, gives

$$\operatorname{Res}(P_4, \pi^{(5)}) = \beta_1 \beta_2^2 \beta_3^3 \beta_4^4 = \frac{n_4^4}{n_0 n_1 n_2 n_3} = \frac{\Psi^4}{\det\big(\nu_i(x^{r+s})\big)_{r,s<4}} = \frac{a\Psi^4}{I_4(i)}.$$

Since $N_0 = I_4(i)$, this is the second equation.

(2) The degree bounds are immediate, since the resultant has total degree at most ten in the nodes. The quadratic part of f is irreducible, with nonzero Hessian determinant. The degree-twelve part of the second equation is $-a\big(\sum_j \gamma_j s_j^3\big)^4$. If f divided it, the quadratic part of f would divide $\sum_j \gamma_j s_j^3$, and an explicit computation excludes this. So the two equations have no common factor, their common zero set is a curve, and by Bézout's theorem its components over $\mathbb{Q}$ have total degree at most $2 \cdot 12$. □

Degrees throughout this section mean the leading coefficient of the eventual affine Hilbert function, equivalently the degree of the projective closure. "Integral" means that the defining ideal is prime over $\mathbb{Q}$; it does not assert geometric irreducibility or a rational parametrization.

15.2. **Reconstruction and retained components.** Two further identities distinguish the components on which the later rational functions have the required behavior. We give their definitions before stating the component bounds.

Form the 8×9 matrix of coefficients of $P_4, xP_4, x^2P_4, x^3P_4, Z, xZ, x^2Z, x^3Z$ in the basis $1, x, \dots, x^8$. Let c_j be $(-1)^j$ times the minor obtained by deleting column j, and define

$$Q = \frac{c_1 - c_2}{c_1 - c_0}, \qquad E_0 = a\,\frac{\det(c_{r+s})_{0\le r,s<3}(c_1 - c_0)^3}{(c_1^2 - c_0c_2)^3}.$$

At a good start the matrix annihilates the moment vector $(h_i(0), \dots, h_i(8))$ and has rank eight. Indeed, the resultant in (13) is positive, so P_4 and Z are coprime. The identities $q_ih_i(0) + h_i(1) = q_ih_i(1) + h_i(2) = 1$, together with $h_i(0)h_i(2) > h_i(1)^2$, then give

$$h_i(j) = \frac{c_j(c_1 - c_0)}{c_1^2 - c_0c_2} \quad (0 \le j \le 8), \qquad Q = q_i, \qquad E_0 = I_3(i).$$

In particular, all displayed denominators are nonzero at good starts.

To state the identities as equations on the curve, use the raw-moment polynomials

$$M_0 = 1, \qquad M_{r+1}(k_0, k_1, \dots) = \sum_{t=0}^{r} \binom{r}{t} k_0^{r-t} M_t(k_1, k_2, \dots).$$

Only $M_0, \dots, M_6$ are needed; they involve $k_0, \dots, k_4$. At good starts, $M_r = \mu_i(x^r)$ for $0 \le r \le 6$, by the weight recursion in Section 10. Set

$$H_3 = \det(M_{r+s})_{0\le r,s<3}, \qquad H_4 = \det(M_{r+s})_{0\le r,s<4}, \qquad H_{013} = \det(M_{r+s})_{\substack{r=0,1,2\\ s=0,1,3}}.$$

This H_4 is the polynomial of Lemma 11.4. With $p_3 = \bar{u}_2 - 1 - s_1 - s_2 - s_3$, the coefficient of x^3 in P_4, the normalization and raw-resolvent identities are

$$N_0P_4(-Q) = aH_4, \qquad E_0P_4(-Q) = a\big((Q - p_3)H_3 - H_{013}\big). \tag{14}$$

Among components containing a good start, we call a component *retained* if both equations, after clearing denominators, hold identically on it.

Lemma 15.2 (Removing components by polynomial equations)**.** *The two identities in* (14)*, after clearing denominators, have degrees at most* 166 *and* 404*, respectively. On a degree-d component, either an identity holds everywhere or it holds at* $166d$ *or* $404d$ *good starts at most, respectively. Consequently each component that is not retained contributes* $O(1)$ *good starts.*

Proof. Write $Q = n_Q/d_Q$ and $E_0 = n_E/d_E$ using the cofactor expressions above. Each cofactor has degree at most 40, so

$$\deg n_Q, \deg d_Q \le 40, \qquad \deg n_E, \deg d_E \le 240.$$

The latter bound follows from the 3×3 determinant and the cube in the numerator, and the cube of a quadratic cofactor expression in the denominator. The polynomial $\widehat{P} = d_Q^4 P_4(-n_Q/d_Q)$ has degree at most 163: in Newton form its longest product is $n_Q \prod_{j=1}^{3}(n_Q + s_j d_Q)$, of degree at most $40 + 3 \cdot 41$, because $s_0 = 0$. The cleared equations are

$$\Phi_N = N_0 \widehat{P} - a H_4 d_Q^4,$$
$$\Phi_R = n_E \widehat{P} - a\big((n_Q - p_3 d_Q) H_3 - H_{013} d_Q\big) d_Q^3 d_E.$$

Here $\deg H_3 \le 3$, $\deg H_4 \le 6$ and $\deg H_{013} \le 4$. Thus $\deg \Phi_N \le 166$ and $\deg \Phi_R \le 404$.

If a polynomial of degree r does not vanish identically on an integral curve of degree d, its zero set on that curve has at most rd distinct rational points, by the polynomial-section form of Bézout's bound. Every good start satisfies both cleared equations. Moreover, the free nodes and pattern recover $q_i = Q$, and q_i strictly increases along the fixed orbit. Thus distinct good starts of this pattern give distinct free-node triples. Applying the point bound to whichever equation is not an identity proves the claim. The absolute bound $d \le 24$ makes the last estimate uniform. □

15.3. **Directions at infinity.** Call a pattern *exceptional* if two of the five Newton coefficients γ_j sum to zero. Since $\bar{N}_1 > 0$, $\bar{u}_1 > 0$ and $\bar{u}_2 < 0$, every γ_j is nonzero, and a pair sum can vanish only in one of the four cases

$$\bar{N}_1 + \bar{u}_2 = 1, \qquad \bar{N}_1 + \bar{u}_2 = 0, \qquad \bar{u}_1 = 1, \qquad \bar{N}_1 + \bar{u}_1 = 1. \tag{15}$$

Equivalently, the integer pattern coordinates satisfy $N_1 + u_2 = N_0$, $N_1 + u_2 = 0$, $u_1 = N_0$ or $N_1 + u_1 = N_0$.

Lemma 15.3 (Leading nodes and determinant orders)**.** *Let C be an integral component of positive degree of* (13)*. Let P be a closed point above infinity on the normalization of the projective closure of C, write* $\mathrm{ord} = \mathrm{ord}_P$*, and let $e > 0$ be the multiplicity of the hyperplane at infinity at P. For a local parameter ϖ at P, let z_j be the residue of $\varpi^e s_j$, $0 \le j \le 4$. Then the z_j are not all equal and satisfy*

$$\sum_{j=0}^{4} \gamma_j z_j^r = 0 \qquad (0 \le r \le 3).$$

If two of these five leading nodes coincide, the pattern is exceptional. For a nonexceptional pattern, therefore,

$$\mathrm{ord}(s_l - s_j) = -e \quad (0 \le j < l \le 4),$$
$$\mathrm{ord}\,\mathrm{Res}(P_4, Z) = -10e,$$
$$\mathrm{ord}\, H_3 = -3e, \qquad \mathrm{ord}\, H_4 = -6e.$$

Positive order denotes a zero and negative order a pole.

Proof. All nonzero rational constants have order zero. The multiplicity e can equivalently be written as

$$e = \max_{1 \le j \le 3} \{-\mathrm{ord}\, s_j\} > 0.$$

Thus a polynomial of total node degree at most r has order at least $-re$. The residues z_j lie in the residue field of P; we do not require P to be rational. We have $z_0 = 0$, and at least one of z_1, z_2, z_3 is nonzero.

The leading nodes are distinct. The definitions of the nodes give the moment equations of orders zero and one. The highest-degree part of $f = 0$ gives the equation of order two. Finally,

the highest-degree part of $N_0 \operatorname{Res}(P_4, Z) = a\Psi^4$ gives the equation of order three: the resultant has total degree at most ten, whereas the cubic part of Ψ is $\sum_j \gamma_j s_j^3$. Consequently

$$\sum_{j=0}^{4} \gamma_j z_j^r = 0 \qquad (0 \le r \le 3). \tag{16}$$

For the last of these, multiply the fixed-dividend equation by ϖ^{12e} and take residues; its left side vanishes and its right side is $a(\sum_j \gamma_j z_j^3)^4$.

If two nodes coincided, there would be at most four distinct values among the z_j. Grouping the sums in (16) by those values and using the corresponding Vandermonde matrix shows that the sum of the γ_j in each group is zero. Since there are five indices and at least two groups, some group has at most two elements. It cannot have one, because every γ_j is nonzero; it must therefore give a vanishing pair sum. This proves the collision assertion. For a nonexceptional pattern it is impossible, so all five nodes are distinct, and

$$\operatorname{ord}(s_l - s_j) = -e \qquad (0 \le j < l \le 4). \tag{17}$$

The determinant orders. Write $\mathcal{R} = \operatorname{Res}(P_4, Z)$. The leading homogeneous parts needed below are

$$[\mathcal{R}]_{10} = \gamma_0\gamma_1\gamma_2\gamma_3 \prod_{0 \le j < l \le 4} (s_l - s_j), \tag{18}$$

$$[H_3]_3 = k_0 k_1 (k_0 + k_1),$$

$$[H_4]_6 = k_0 k_1 k_2 (k_0 + k_1)(k_1 + k_2)(k_0 + k_1 + k_2). \tag{19}$$

Here $[F]_r$ denotes the homogeneous part of total node degree r. For completeness, the degree-ten formula can be checked without expanding the resultant. In the Newton basis beginning at s_1, s_2, s_3, its 4×4 Bezout matrix for P_4, R_4 has entry degrees at most $4 - \max(i, j)$, for $0 \le i, j < 4$. Only the diagonal term of its determinant can have degree ten. Its four leading diagonal entries are

$$\begin{aligned} &-\gamma_1 s_1 (s_1 - s_2)(s_1 - s_3)(s_1 - s_4), \\ &-\gamma_2 s_2 (s_2 - s_3)(s_2 - s_4), \\ &-\gamma_3 s_3 (s_3 - s_4), \qquad -s_4. \end{aligned}$$

Their product gives the stated Vandermonde; the additional factor x in the second polynomial contributes $P_4(0) = \gamma_0$. The raw-moment recurrence gives

$$H_3 = k_0 k_1 (k_0 + k_1) + k_0 k_2 - k_1^2,$$

and gives the displayed degree-six part of H_4, with remainder of degree at most five. These are finite polynomial identities.

Every factor in these leading products has order $-e$ by (17). The lower-degree terms have strictly larger order, so they cannot cancel the leading terms. Thus

$$\operatorname{ord} \mathcal{R} = -10e, \qquad \operatorname{ord} H_3 = -3e, \qquad \operatorname{ord} H_4 = -6e. \tag{20}$$

□

Corollary 15.4 (Even component degrees)**.** *Every positive-degree nonexceptional component of* (13) *has even degree. At each point above infinity its hyperplane multiplicity e is even, and* $\operatorname{ord}(N_0\Psi) = -5e/2$.

Proof. Taking orders in the fixed-dividend equation and using Lemma 15.3 gives $4 \operatorname{ord} \Psi = -10e$. Since orders are integers and $N_0 \neq 0$, this proves both local assertions. The degree of the curve is the sum of the multiplicities e, weighted by the residue-field degrees, so it is even. The retained identities are not needed here. □

15.4. **Exceptional patterns and line components.** Exceptional patterns may have components of odd degree, including lines, but there are fewer of them.

Lemma 15.5 (Exceptional patterns and lines)**.** *There are $O(X^4 \log(2X)) = O_\eta(a^\eta X^4)$ exceptional patterns. In the range*

$$124928A^4H^5X < a^3, \tag{21}$$

no line component contains a good start.

Proof. Each relation in (15) becomes a vanishing maximal minor after a fixed invertible integer column change. The transformed minor vector still has height $O(X)$. For example, if the original columns are $C_0, \dots, C_4$, then

$$\det(C_1, C_2 + C_0, C_3 - C_0) = p_{123} - p_{013} - p_{012} = N_1 + u_2 - N_0.$$

The other three relations follow by the same column additions and permutations.

A fixed zero minor reduces the Plücker relations to three integer vectors $\mathbf{x}, \mathbf{y}, \mathbf{z}$ of height $O(X)$ with $\mathbf{x} \times \mathbf{y} = 0$ and $\mathbf{x} \cdot \mathbf{z} = \mathbf{y} \cdot \mathbf{z} = 0$. These vectors are signed rearrangements of the nine remaining minors, so they determine the minor vector. The case $\mathbf{x} = \mathbf{y} = 0$ has $O(X^3)$ choices. Otherwise write $\mathbf{x} = r\mathbf{v}$, $\mathbf{y} = s\mathbf{v}$ with $\mathbf{v}$ primitive. In a dyadic range $R \leq |\mathbf{v}|_\infty < 2R$, there are $O(R^3)$ choices for $\mathbf{v}$ and $O((X/R)^2)$ for (r, s). The equation $\mathbf{v} \cdot \mathbf{z} = 0$ has $O(X^2/R)$ solutions in the box. Indeed, take a coordinate of absolute value $n = |\mathbf{v}|_\infty$. Primitivity gives n admissible pairs of the other two coordinates modulo n, and the chosen coordinate is then determined. Since $R \leq n \ll X$, this gives $O(X^2/n)$ choices. Each range therefore contributes $O(X^4)$, and summing over $O(\log(2X))$ ranges proves the pattern count. The crude bound $X \leq 10^{16}a^7$ absorbs its logarithm into a^η.

For a contained line, the degree-twelve part of (13) first forces the cubic coefficient of Ψ to vanish. Then Ψ^4 has degree at most eight, so the degree-ten resultant coefficient also vanishes. By the leading forms and collision argument in Lemma 15.3, its direction satisfies one of the four exceptional relations. The grouping argument there gives exactly two groups of equal leading nodes, of sizes two and three. After scaling, the five-node directions are, in the order of (15),

$$(0,1,1,0,1), \quad (0,1,1,0,0), \quad (0,0,1,1,0), \quad (0,1,0,0,1).$$

Substitution of these directions into the resultant gives a polynomial of degree at most six. Equation (13) then forces Ψ to have degree at most one along the line. Its quadratic coefficient, together with $f = 0$, gives respectively

$$s_3 = 0, \qquad k_1 = 1, \qquad k_2 = 1, \qquad s_4 - s_1 = \bar{u}_1 - \bar{u}_2.$$

The first is impossible at a good start. The polynomial G in Lemma 11.5 satisfies, by expansion,

$$G \leq H^5(12k_1 + 464), \qquad G \leq H^5(12k_2 + 464).$$

In the last obstruction, $k_1 + k_2 + k_3 = \bar{u}_1 - \bar{u}_2 = (u_1 - u_2)/N_0 \leq 2X$. Thus each surviving case gives $k_1 \leq 2X$ or $k_2 \leq 2X$. The label bound $N_0 q_i^4 \leq aG$, with $N_0 \geq 1$, now gives

$$q_i^4 \leq aH^5(24X + 464).$$

But $q_i \geq a/(4A)$ and $24X + 464 \leq 488X$ for $X \geq 1$. Condition (21), with $124928 = 4^4 \cdot 488$, gives the strict reverse inequality. □

15.5. **Orders of the reconstructed functions.**

Lemma 15.6 (Orders of the reconstructed functions)**.** *Let C be a retained nonexceptional component of positive degree containing a good start. At every point above infinity, with hyperplane multiplicity e, one has*

$$\operatorname{ord} Q = -\frac{3e}{2}, \qquad \operatorname{ord} E_0 = \frac{3e}{2}.$$

In particular, E_0 is a nonzero rational function vanishing at every point above infinity.

Proof. The good start guarantees that the reconstruction denominators are not identically zero on C. Thus Q and E_0 are rational functions on C, and retention makes both identities (14) identities in its function field.

The quotient order. Write $P_4(x) = x^4 + p_3x^3 + p_2x^2 + p_1x + p_0$. Its Newton expression gives $\deg p_j \le 4 - j$, and hence $\operatorname{ord} p_j \ge -(4-j)e$. The first retained identity gives

$$\operatorname{ord} P_4(-Q) = \operatorname{ord}(aH_4/N_0) = -6e.$$

If $\operatorname{ord} Q \ge -e$, every term of $P_4(-Q)$ would have order at least $-4e$, a contradiction. Put $r = \operatorname{ord} Q < -e$. For $0 \le j < 4$,

$$\operatorname{ord}\big(p_j(-Q)^j\big) - 4r \;\ge\; (4-j)(-e-r) > 0.$$

Thus Q^4 is the unique term of lowest order. It follows that $4r = -6e$, or $\operatorname{ord} Q = -3e/2$.

The reconstructed curvature order. The coefficient $p_3 = \bar{u}_2 - 1 - s_1 - s_2 - s_3$ has order at least $-e$. The raw-moment recurrence also shows that H_{013} has degree at most four: its degree-four part is $(2k_0 + k_1)k_0k_1(k_0 + k_1)$. Consequently

$$\operatorname{ord}(Q - p_3) = -3e/2, \qquad \operatorname{ord}\big((Q-p_3)H_3\big) = -9e/2 < -4e \le \operatorname{ord} H_{013}.$$

There is again a unique term of lowest order. Dividing the second retained identity by $P_4(-Q)$ therefore gives

$$\operatorname{ord} E_0 = -9e/2 - (-6e) = 3e/2.$$

The argument uses the discrete valuation on the normalization, so it also covers branches above singular points of the original curve. □

15.6. **The joint coordinates.** Put $d_j = b_{i+j} - b_{i+j+1}$ and

$$e_{01} = d_0k_1 - d_1k_0, \quad e_{02} = d_0k_2 - d_2k_0, \quad e_{12} = d_1k_2 - d_2k_1.$$

The first of these, e_{01}, is the curvature $I_3(i)$. With $s_j = q_{i+j} - q_i$ and $r_j = k_j(s_j + s_{j+1}) + k_{j+1}$, write

$$B_0 = k_0r_1 - k_1r_0 = k_0^2k_1 + (k_0 - 1)k_1^2 + k_0k_2,$$
$$B_1 = k_1r_2 - k_2r_1 = k_1^2k_2 + (k_1 - 1)k_2^2 + k_1k_3.$$

They are positive at every state at which they are defined, and $B_0 = H_3$. On a retained component, use the reconstructed function E_0 to define

$$E_{12} = \frac{B_1E_0 - N_0k_1}{B_0}, \qquad E_{02} = \frac{k_0E_{12} + k_2E_0}{k_1}, \qquad W = (E_{12}, -E_{02}, E_0), \qquad J = N_0\Psi.$$

The determinant and cross-product identities give, identically,

$$W \cdot (k_0, k_1, k_2) = 0, \qquad W \cdot (r_0, r_1, r_2) = N_0. \tag{22}$$

At a good start these rational functions take the integer values $W = (e_{12}, -e_{02}, e_{01})$. Changing the rank-five Gram basis to $1, x, x^2, x^3, P_4$ gives $\det(\nu_i(x^{r+s}))_{r,s<5} = \det(\nu_i(x^{r+s}))_{r,s<4}\nu_i(P_4^2)$.

Consequently $J = I_5(i)$ is a positive integer at good starts. The heights of these integers are controlled by

$$S = \left\lfloor \frac{4352A^3H^3}{a^2} \right\rfloor, \qquad Y = \left\lfloor \frac{1259991040A^5H^{10}}{a^4} \right\rfloor.$$

Lemma 15.7. *At good starts, $|W_j| \le S$ and $1 \le J \le Y$.*

Proof. Write $q = q_i$ and use the exact positive remainder expansion $(q+y)^{-1} = q^{-1} - yq^{-2} + y^2/(q^2(q+y))$. For the first four shifted states,

$$\frac{b_{i+j}}{a} = \frac{1}{q} - \frac{s_j + 1}{q^2} + \epsilon_j, \qquad 0 \le \epsilon_j \le 17H^2/q^3.$$

Thus $d_j/a = k_j/q^2 + O(34H^2/q^3)$, and cancellation of the leading terms in each minor gives $|e_{jl}|q^3 \le 68aH^3$. The bound $q \ge a/(4A)$ gives S, including the floor because the minors are integers. For J, Newton basis Gram comparison bounds the raw rank-five determinant by $(1 \cdot 2 \cdot 10 \cdot 77 \cdot 799)H^{10} = 1230460H^{10}$. Weighted Gram comparison gives $Jq^5 \le 1230460aH^{10}$; the same lower bound for q gives Y. □

Corollary 15.8 (Orders of the joint coordinates)**.** *In the setting of Lemma 15.6, the joint coordinates satisfy*

$$\operatorname{ord} W_0 = \operatorname{ord} W_2 = 3e/2, \qquad \operatorname{ord} W_1 \ge 3e/2, \qquad \operatorname{ord} J = -5e/2. \tag{23}$$

Proof. The cubic leading part of B_1 is $k_1k_2(k_1+k_2)$. By (17), it has order $-3e$, whereas the quadratic remainder $k_1k_3 - k_2^2$ has order at least $-2e$. Thus $\operatorname{ord} B_0 = \operatorname{ord} B_1 = -3e$ and $\operatorname{ord} k_1 = -e$. In the numerator defining E_{12} the two terms have different orders:

$$\operatorname{ord}(B_1E_0) = -3e/2 < -e = \operatorname{ord}(N_0k_1).$$

Therefore $\operatorname{ord} E_{12} = 3e/2$. Each term in $k_0E_{12} + k_2E_0$ has order $e/2$, so their sum has order at least $e/2$. Dividing by k_1 gives $\operatorname{ord} E_{02} \ge 3e/2$. Cancellation here is harmless: equality for the first and third coordinates already gives $\min_j \operatorname{ord} W_j = 3e/2$. The formula for $J = N_0\Psi$ is Corollary 15.4. □

15.7. **Counting the nonexceptional conics.** We now show that a nonexceptional degree-two component contributes only $O_\eta(a^\eta)$ good starts. The implied constant is independent of the pattern and of the component. A component that is not retained already contributes at most $404 \cdot 2 = 808$ starts, by Lemma 15.2. We may therefore assume that the component C is retained. If it contains at most two free-node triples of good starts, there is nothing to prove. In the remaining case we first construct a parameter whose value is an integer at every such triple.

Lemma 15.9 (A bounded integral parameter)**.** *Let C be a retained nonexceptional degree-two component containing the free-node triples of at least three good starts. After permuting the three free-node coordinates to (x, y, z), there are an integer $\Delta_c \ne 0$ and polynomials $\xi_0, \xi_1, \xi_2 \in \mathbb{Z}[t]$ such that*

$$\deg \xi_j \le 2, \qquad |\Delta_c|,\ \max_{j,k} |[t^k]\xi_j| \le (aX)^{O(1)}.$$

The map

$$t \longmapsto \big(\xi_0(t), \xi_1(t), \xi_2(t)\big)/\Delta_c$$

parametrizes C over $\mathbb{Q}$. Its inverse takes an integer value at every integer point of C, and is injective on those points.

Proof. Every free node of a good start has absolute value at most a. Three distinct such triples admit an integer plane

$$\lambda_z z + \lambda_x x + \lambda_y y + \lambda_1 = 0, \qquad |\lambda_z|, |\lambda_x|, |\lambda_y| \le 8a^2, \qquad |\lambda_1| \le 24a^3.$$

For example, take a cross product of two differences to obtain the normal; if that cross product vanishes, a nonzero perpendicular vector with the same bounds suffices. Put $K_{\mathrm{p}} = 24a^3$. A plane not containing an integral curve of degree two meets it in at most two distinct points. Thus this plane contains C. Choose the eliminated coordinate z with $\lambda_z \neq 0$.

The six rational pattern coordinates (12) have common denominator $N_0 N_1 \neq 0$ and respective integer numerators

$$N_1^2,\ u_1 N_1,\ u_2 N_1,\ N_2 N_0,\ v_1 N_0,\ v_2 N_0,$$

each of absolute value at most X^2. Expanding the quadratic f, its homogeneous part is a diagonal quadratic plus the square of a linear form. Its three diagonal numerators have denominator $N_0 N_1$, and its three linear numerators have denominator $(N_0 N_1)^2$; all six numerators have absolute value at most $12X^4$. Substitute $z = -(\lambda_x x + \lambda_y y + \lambda_1)/\lambda_z$ and multiply by $(N_0 N_1 \lambda_z)^2$. The resulting plane equation is

$$\alpha x^2 + \beta xy + \gamma_c y^2 + d_c x + e_c y + f_c = 0, \tag{24}$$

with integer coefficients of absolute value at most $K = 10(12X^4)^2 K_{\mathrm{p}}^2$. Indeed, each coefficient is a sum of products of two quantities bounded by $12X^4$ and two plane coefficients bounded by K_{p}; expansion gives the stated factor 10. The restriction of f to the plane is nonzero: otherwise the plane would be a component of the irreducible pattern quadric. It has degree two because it contains C, and hence defines C in that plane.

The pullback of the hyperplane at infinity to the normalization of C has degree two. Corollary 15.4 says that each of its positive multiplicities is even. Counting residue degrees, there is therefore just one branch over infinity, of multiplicity two and residue degree one. Consequently the leading binary quadratic in (24) is a nonzero rational scalar times the square of a rational linear form. Indeed, the normalization map is finite and surjective, so the projective conic meets the line at infinity only at the image of this rational point. Its leading binary quadratic therefore has a single rational zero, of multiplicity two. Interchanging x and y if necessary gives

$$\alpha \neq 0, \qquad \beta^2 - 4\alpha\gamma_c = 0.$$

Set

$$\kappa_c = 2\beta d_c - 4\alpha e_c, \qquad \epsilon_c = d_c^2 - 4\alpha f_c, \qquad t = 2\alpha x + \beta y + d_c.$$

Completing the square in (24) gives

$$t^2 = \kappa_c y + \epsilon_c. \tag{25}$$

Here $\kappa_c \neq 0$. If it were zero, a rational point of C with parameter t_0 would give $\epsilon_c = t_0^2$, and the equation would factor as $(t - t_0)(t + t_0) = 0$. Integrality would then place C in an affine line, contradicting its degree.

We can therefore solve explicitly:

$$y = \frac{t^2 - \epsilon_c}{\kappa_c}, \qquad x = \frac{t - \beta y - d_c}{2\alpha}, \qquad z = -\frac{\lambda_x x + \lambda_y y + \lambda_1}{\lambda_z}.$$

Substitution verifies the conic and plane equations for every rational t, and the displayed linear expression in x, y recovers t. In particular, t is an integer whenever (x, y, z) is an integer point.

Distinct points have distinct parameters. Taking $\Delta_c = 2\alpha\kappa_c\lambda_z$ clears all three denominators. The inequalities

$$|\kappa_c| \le 6K^2, \qquad |\epsilon_c| \le 5K^2, \qquad |\beta\epsilon_c - d_c\kappa_c| \le 11K^3$$

give $|\Delta_c| \le 12K_{\mathrm{p}}K^3$ and the coefficient bounds $33K_{\mathrm{p}}K^3$ for the three resulting quadratic numerators. For good starts one also has $|t| \le 4Ka$, although the counting argument below does not need a parameter interval. □

Lemma 15.10 (Properness and coefficient height)**.** *In the parameter of Lemma 15.9, write*

$$E_0(t) = \frac{F_E(t)}{G_E(t)}, \qquad F_E, G_E \in \mathbb{Z}[t], \qquad \gcd_{\mathbb{Q}[t]}(F_E, G_E) = 1.$$

The polynomials may be chosen so that

$$\deg G_E - \deg F_E = 3, \qquad \max(\deg F_E, \deg G_E) \le D_0, \qquad \max_k\big(|[t^k]F_E|, |[t^k]G_E|\big) \le C_0 a^{k_0},$$

for absolute constants D_0, C_0, k_0. In particular, G_E is nonconstant. At the parameter t of every good start, $G_E(t) \neq 0$ and $F_E(t)/G_E(t) = I_3(i)$ is an integer.

Proof. The explicit parametrization identifies C with the affine line and its smooth projective model with $\mathbb{P}^1$. Its point at infinity has hyperplane multiplicity two. Lemma 15.6 therefore gives

$$\operatorname{ord}_{t=\infty} E_0 = 3.$$

The function is nonzero, since it is positive at a good start. For a reduced rational function on $\mathbb{P}^1$, its order at infinity is $\deg G_E - \deg F_E$, proving properness.

We next bound the coefficients uniformly in the component, including after cancellation. Retain $K_{\mathrm{p}} = 24a^3$ and $K = 10(12X^4)^2K_{\mathrm{p}}^2$ from the preceding proof, and define

$$K_{\mathrm{h}} = 1 + a + X + 33K_{\mathrm{p}}K^3.$$

The six rational pattern coordinates, the three coordinate polynomials ξ_j/Δ_c, and the fixed dividend a have the common nonzero denominator $N_0N_1\Delta_c$. Their ten numerator polynomials have degree at most two and coefficient height at most K_{h}^4; the common denominator has absolute value at most K_{h}^4 as well. For instance, the coordinate numerators are $N_0N_1\xi_j$, while the numerator for a is $aN_0N_1\Delta_c$.

The cofactor formula defining E_0 is a quotient of two fixed polynomials in these ten inputs, with fixed rational coefficients. Clear those fixed coefficients once, then substitute the ten polynomials and clear a fixed power of their common denominator. This gives integer polynomials $\tilde{n}, \tilde{d}$, both nonzero, of bounded degree and coefficient height at most CK_{h}^r, where the degree bound, C and r are absolute. This assertion follows directly from polynomial multiplication: both the number of factors and the number of summands are fixed by the displayed cofactor formula.

Choose a primitive $\Pi \in \mathbb{Z}[t]$ representing $\gcd_{\mathbb{Q}[t]}(\tilde{n}, \tilde{d})$. By Gauss's lemma, Π divides both $\tilde{n}$ and $\tilde{d}$ in $\mathbb{Z}[t]$. Thus $F_E = \tilde{n}/\Pi$ and $G_E = \tilde{d}/\Pi$ are integer polynomials with the same ratio, coprime over $\mathbb{Q}$, and with degrees no larger than those of $\tilde{n}, \tilde{d}$. For a nonzero integer polynomial of degree at most D_0 and coefficient height $\mathcal{H}$, every integer polynomial factor has coefficient height at most $2^{D_0}(D_0+1)\mathcal{H}$. To see this, the Mahler measure of an integer cofactor is at least one, the original polynomial's Mahler measure is at most $(D_0+1)\mathcal{H}$, and each coefficient of a factor is bounded by 2^{D_0} times its Mahler measure. Thus cancellation preserves a fixed polynomial bound in K_{h}.

Finally, $K_{\mathrm{p}} = 24a^3$ and $X \le 10^{16}a^7$ show that K_{h} is bounded by a fixed power of a. Absorbing the fixed exponents and constants gives the asserted $C_0a^{k_0}$ bound. The original

reconstruction denominator is nonzero at every good start. Its cleared polynomial $\tilde{d}$ is therefore nonzero at the corresponding parameter, and so is its factor G_E. The reconstruction identity then gives $F_E(t)/G_E(t) = I_3(i) \in \mathbb{Z}$. □

Lemma 15.11 (Integer values of a rational function)**.** *Let $F_E, G_E \in \mathbb{Z}[t]$ be coprime over $\mathbb{Q}$, of degrees at most D_0, with G_E nonconstant and coefficient heights at most $C_0 a^{k_0}$. For every $\eta > 0$, the number of integers t such that $G_E(t) \neq 0$ and $F_E(t)/G_E(t) \in \mathbb{Z}$ is $O_\eta(a^\eta)$. The implied constant depends only on η, D_0, C_0, k_0.*

Proof. The integer resultant $R = \operatorname{Res}(F_E, G_E)$ is nonzero. The adjugate of the Sylvester matrix gives $U, V \in \mathbb{Z}[t]$ with $UF_E + VG_E = R$. At any parameter in question, $G_E(t) \mid F_E(t)$, and hence $G_E(t) \mid R$. There are $2\tau(|R|)$ possible signed values of $G_E(t)$. For each such value v, the nonzero polynomial $G_E - v$ has at most $\deg G_E \leq D_0$ roots. Thus the number of parameters is at most

$$2D_0\tau(|R|).$$

The Sylvester matrix has size at most $2D_0$, and its entries are bounded by $C_0 a^{k_0}$. Expanding its determinant gives

$$1 \leq |R| \leq (2D_0)!\big(C_0 a^{k_0}\big)^{2D_0}.$$

The divisor estimate $\tau(n) \ll_\varepsilon n^\varepsilon$, with ε chosen in terms of η and the fixed degree and height exponents, proves the claim. □

Apply the last lemma to the reconstructed invariant E_0. The parameter is injective on free-node triples, and reconstruction of the strictly increasing q_i makes those triples injective in the good starts. Each retained nonexceptional conic therefore contributes $O_\eta(a^\eta)$ starts. The components with at most two free-node triples of good starts and the nonretained components were already bounded absolutely. Apply the conic estimate with $\eta/2$ and the pattern estimate $O_{\eta/2}(a^{\eta/2}X^5)$. Since there are at most twenty-four components per pattern, all nonexceptional conics together contribute $O_\eta(a^\eta X^5)$.

15.8. **The extra degree and the powered graph.** The conic count leaves retained nonexceptional components of degree at least four. Fix such a component C of degree d. The additional integer coordinates will increase its degree without increasing their heights too much. The key is one extra zero over the affine part of C.

All divisors in this argument are on the normalization $\widetilde{C}$ of the projective closure of C, over $\mathbb{Q}$. For a closed point P of $\widetilde{C}$, put $f_P = [\kappa(P) : \mathbb{Q}]$, where $\kappa(P)$ is the residue field; thus $\deg(\sum_P a_P[P]) = \sum_P a_P f_P$. Let $E_\infty = \sum_{P|\infty} e_P[P]$ be the pullback of the original hyperplane at infinity. Then $\sum_{P|\infty} e_P f_P = d$. We do not require the closed points, or the branches over a rational point of C, to be rational.

Let $\mathcal{F}$ be the pole envelope of the curvature coordinates:

$$\mathcal{F} = \sum_P \max\{0, -\operatorname{ord}_P(W_0), -\operatorname{ord}_P(W_1), -\operatorname{ord}_P(W_2)\}[P],$$

with $\operatorname{ord}_P(0) = +\infty$. The orders in (23) show that $\mathcal{F}$ is supported over the affine part of C, and hence is disjoint from E_∞.

Lemma 15.12 (An additional affine zero)**.** *There is a rational linear combination ϕ of W_0, W_1, W_2 which is not the zero function and has a positive-order zero at a closed point P_0 of $\widetilde{C}$ outside E_∞.*

Proof. First, the W_j cannot all be scalar multiples, in $\mathbb{Q}(C)$, of a fixed nonzero vector $v \in \mathbb{Q}^3$. Such a relation, together with $W \cdot (k_0, k_1, k_2) = 0$, would put C in the plane $v_0k_0+v_1k_1+v_2k_2 = 0$. This is a genuine plane: the first three gaps are an invertible linear change of the three node coordinates. The pattern quadratic is irreducible, so its restriction to this plane is nonzero and has degree at most two. Every curve contained in this plane section therefore has degree at most two, contrary to $d \geq 4$.

Choose the free-node triple $z_* \in C(\mathbb{Q})$ of a good start. All the W_j are regular there, because their common reconstruction denominator does not vanish at a good start. Write $w_j = W_j(z_*)$. The identity $W \cdot (r_0, r_1, r_2) = N_0 \neq 0$ gives $w \neq 0$. Choose h with $w_h \neq 0$. At least one function $w_hW_j - w_jW_h$ is nonzero: otherwise $W = (W_h/w_h)w$, contradicting the preceding paragraph. Take such a function for ϕ. It has rational coefficients and satisfies $\phi(z_*) = 0$.

This zero survives normalization even if z_* is singular. Indeed, write $\phi = g/h_0$ near z_*, with g not identically zero on C, $g(z_*) = 0$ and $h_0(z_*) \neq 0$. The finite normalization has a closed point P_0 above z_*. In its discrete valuation ring the numerator is in the maximal ideal and the denominator is a unit, so $\operatorname{ord}_{P_0}(\phi) > 0$. Since z_* is affine, P_0 does not lie in E_∞. □

Lemma 15.13 (The curvature pole degree)**.** *The pole envelope satisfies*

$$\deg \mathcal{F} \geq \frac{3d}{2} + 1. \tag{26}$$

Proof. Use the function ϕ from Lemma 15.12. At every point over infinity, each W_j has order at least $3e_P/2$. The same is true of a rational linear combination; cancellation can only increase its zero order. Consequently

$$(\phi)_0 \;\geq\; \frac{3}{2}E_\infty + [P_0], \qquad (\phi)_\infty \;\leq\; \mathcal{F}.$$

The first divisor is integral because every e_P is even. The function ϕ is nonzero and has a zero, so its principal divisor has equal zero and pole degrees. Hence

$$\deg \mathcal{F} \;\geq\; \deg(\phi)_\infty = \deg(\phi)_0 \;\geq\; \frac{3}{2}\sum_{P|\infty} e_P f_P + f_{P_0} \;\geq\; \frac{3d}{2} + 1.$$

The last inequality uses only $f_{P_0} \geq 1$, not that P_0 is rational. □

For positive integers m, n, let $\Gamma_{m,n} \subseteq \mathbb{A}^7$ be the closure of the graph obtained by adjoining W_0^m, W_1^m, W_2^m, J^n to the three original node coordinates. The graph is taken on the common domain of these rational functions. It is an integral curve, and projection to the first three coordinates is birational: on that domain the four remaining coordinates are uniquely determined. Thus $\Gamma_{m,n}$ and C have the same function field. Taking powers does not lose this property, because the original coordinates remain in the graph.

Lemma 15.14 (Degree of the powered graph)**.** *If $\delta = \deg \Gamma_{m,n}$, then*

$$2\delta \geq (5n + 3m)d + 2m, \qquad \delta \geq 10n + 7m =: \delta_{\min}. \tag{27}$$

Moreover,

$$\delta \leq \big(247(m + n) + 1\big)d.$$

For $m = n = 1$ one has the sharper upper bound $\delta \leq 247d$ and the lower bound $\delta \geq 4d+1 \geq 17$.

Proof. Lower bound. At a point P over the old infinity hyperplane, the original node coordinates have pole orders at most e_P, while J^n has pole order $5ne_P/2$ and all W_j^m vanish. Since $n \geq 1$, the pole envelope of the seven graph coordinates at P is exactly $5ne_P/2$. Over the affine part, the node coordinates and J are regular: $J = N_0\Psi$ is polynomial in the nodes. There

the envelope is exactly m times the curvature envelope. The complete envelope of the graph coordinates is therefore

$$\mathcal{F}_{m,n} = \frac{5n}{2} E_\infty + m\mathcal{F}.$$

There is a single rational linear combination λ of the seven coordinates whose pole divisor is $\mathcal{F}_{m,n}$. To see this, fix a point P in the finite support of $\mathcal{F}_{m,n}$ and multiply all seven coordinates by a local uniformizer to the power $\mathcal{F}_{m,n}(P)$. Their residues in $\kappa(P)$ are not all zero. Cancellation of the leading pole is therefore a proper $\mathbb{Q}$-linear condition on the seven coefficients. The finitely many such proper subspaces do not cover $\mathbb{Q}^7$. Choosing coefficients outside their union gives the asserted simultaneous absence of cancellation, also when $\kappa(P) \neq \mathbb{Q}$.

As a linear function on $\Gamma_{m,n}$, λ has degree at most δ. For example, homogenize it on the projective closure of the graph: its pencil with the homogenizing coordinate consists of hyperplane sections of degree δ. Removing common zeros can only decrease the degree of the resulting map to $\mathbb{P}^1$. Birational projection identifies its function field with $\mathbb{Q}(C)$, so this degree is $\deg(\lambda)_\infty = \deg \mathcal{F}_{m,n}$. It follows that

$$\delta \;\geq\; \frac{5n}{2} d + m \deg \mathcal{F} \;\geq\; \frac{5n+3m}{2} d + m.$$

This is the first inequality in (27); $d \geq 4$ gives the second. The auxiliary linear combination is used only to measure degree: no bound on its coefficients enters the integer-point box.

Upper bound. Here only total polynomial degrees matter, not the heights of the pattern-dependent coefficients. Write the cofactor formula for the reconstructed invariant as $E_0 = n_E/d_E$, as in the proof of Lemma 15.2. Each entry of the moment matrix has degree at most four in the nodes. Expanding its 8×8 cofactors gives, in particular, the uniform bound $\deg c_j \leq 40$, so $\deg n_E, \deg d_E \leq 240$. Since B_0, B_1 have degree at most three and the gaps have degree at most one, the three curvature coordinates have a common denominator

$$\Omega = d_E B_0 k_1, \qquad \deg \Omega \leq 244,$$

and numerators $\mathcal{N}_0, \mathcal{N}_1, \mathcal{N}_2$ of degree at most 244. The numerator for J with this denominator is $\mathcal{N}_3 = J\Omega$, of degree at most 247. The denominator is nonzero on C, since it is nonzero at a good start.

Put $\bar{m} = m + n$. Represent the powered coordinates with denominator $\Omega^{\bar{m}}$ and numerators

$$\mathcal{N}_j^m \Omega^{\bar{m}-m} \quad (0 \leq j \leq 2), \qquad \mathcal{N}_3^n \Omega^{\bar{m}-n}.$$

All these numerators have degree at most $247\bar{m}$. The original node coordinates, written with the same denominator, have numerators of degree at most $244\bar{m}+1$. It follows that multiplying a polynomial of degree at most t in the graph coordinates by $\Omega^{\bar{m}t}$ gives a polynomial in the original nodes of degree at most $(247\bar{m}+1)t$. This operation is injective on the corresponding spaces of functions, because Ω is nonzero in $\mathbb{Q}(C)$. Comparing their eventual affine Hilbert functions gives

$$\delta t + O(1) \leq d(247\bar{m}+1)t + O(1),$$

and hence the asserted upper bound. In the unpowered case use Ω directly: the four new numerators have degree at most 247, and the original coordinates have numerators of degree at most 245. The same comparison gives $\delta \leq 247d$. □

Every good start whose free-node triple lies on C gives an integer point on this graph. By Lemma 15.7, its coordinates have absolute values at most

$$\max(3H, S^m, Y^n).$$

The original node coordinates also show that distinct such node triples remain distinct after adjoining the powers.

15.9. **The window bound and the final exponents.** We use the coefficient-uniform affine-curve theorem of Castryck–Cluckers–Dittmann–Nguyen [3, Theorem 3], also stated as [4, Corollary 2.2]: an integral affine curve C over $\mathbb{Q}$ of degree δ in $\mathbb{A}^r$, with $r \geq 2$ fixed, has

$$N_{\mathrm{aff}}(C, B) \ll_r \delta^3 B^{1/\delta}(\log B + \delta), \qquad B \geq 1, \tag{28}$$

integer points in the box $[-B, B]^r$, with an implied constant independent of the coefficients of C. Its formal proof is discussed in Section 16.

Proposition 15.15 (Window bound)**.** *Fix positive integers m, n and $\eta > 0$, and put $\mathcal{B} = \max(H, S^m, Y^n)$. For a window run in $[A, 2A)$ with $T \geq 36$ satisfying* (21)*,*

$$T \ll_{m,n,\eta} a^{\eta}\left(X^5 + X^4 H^{1/2} + X^5 \mathcal{B}^{1/\delta_{\min}}\right). \tag{29}$$

Proof. At least $T/2$ starts are good, and for each pattern the free-node triples of its good starts are distinct (Lemma 15.2). By Lemma 15.1, these triples lie on at most twenty-four integral curves of degree at most twenty-four, and we bound the good starts on each curve, allowing overlaps. All the heights H, S, Y and X are at most a fixed power of a, so the logarithm in (28) is $O_\eta(a^\eta)$.

- *Exceptional patterns.* There are $O_\eta(a^\eta X^4)$ of them (Lemma 15.5). Under (21) no line contains a good start, so every relevant curve has degree $2 \leq \delta \leq 24$, and (28) in $\mathbb{A}^3$ with $B = 3H$ gives $O_\eta(a^\eta H^{1/2})$ good starts per pattern. This gives the term $X^4 H^{1/2}$.
- *Nonexceptional patterns: components that are not retained, and conics.* By Lemma 15.2 and Section 15.7, these contribute $O_\eta(a^\eta)$ per pattern, hence $O_\eta(a^\eta X^5)$ in all.
- *Nonexceptional patterns: the remaining components.* Lines do not occur, by Lemma 15.5, and odd degrees do not occur, by Corollary 15.4, so these components have degree $d \geq 4$. Their good starts give distinct integer points of $\Gamma_{m,n} \subseteq \mathbb{A}^7$ in a box of side $\max(3H, S^m, Y^n) \leq 3\mathcal{B}$. By Lemma 15.14, the degree δ of $\Gamma_{m,n}$ satisfies $\delta_{\min} \leq \delta \leq 24(247(m+n)+1)$, so (28) gives $O_{m,n,\eta}(a^\eta \mathcal{B}^{1/\delta_{\min}})$ good starts per component. This gives the term $X^5 \mathcal{B}^{1/\delta_{\min}}$.

The added coordinates W_j and J are functions of the free nodes on each component. They therefore introduce no further freely counted coordinates, and the pattern count is not enlarged. □

Corollary 15.16. *For fixed positive integers m, n and every $\varepsilon > 0$,*

$$P(a, b) \ll_{m,n,\varepsilon} a^{E(m,n)+\varepsilon}, \qquad E(m, n) = \max\left(\frac{15\delta_{\min} + m}{51\delta_{\min} + 3m},\ \frac{15\delta_{\min} + 6n}{51\delta_{\min} + 20n}\right),$$

where $\delta_{\min} = 10n + 7m$. In particular $E(1, 1) = 128/435$ and $E(50, 51) = 2201/7480$.

Proof. We bound the number T of terms in each dyadic window, eliminating the window scale A. If $T < 36$ there is nothing to prove. If $AT > 28a$, then Proposition 7.3 gives $T^4 \ll a$. Otherwise

$$HAT \leq 56a, \quad Xa^3 \leq 3609600A^4H^6, \quad T^2 \leq 8\Delta A,$$

the last by Proposition 5.1. If (21) fails, these give $T^{17} \ll \Delta^3 a^5$. If (21) holds, we use Proposition 15.15. The first two terms of (29) give $T^{51} \ll a^{15+O(\eta)}$ and $T^{85} \ll a^{25+O(\eta)}$, after absorbing divisor factors. All these exponents equal $5/17$ up to the arbitrarily small loss.

Suppose the graph term dominates. Raising the bound to the power $\delta_{\min}$ gives $T^{\delta_{\min}} \leq KX^{5\delta_{\min}}\mathcal{B}$, where $K \ll_{m,n,\eta} a^{\eta\delta_{\min}}$ includes the counting loss. Suppose also that $\mathcal{B}T^{e_T} \leq$

$C_1 a^{e_a}$, where C_1 may include divisor factors. Raising these inequalities to suitable powers and cancelling positive powers of A and a gives

$$T^{51\delta_{\min}+e_T} \le K\, 3609600^{5\delta_{\min}} 56^{30\delta_{\min}} C_1 (8\Delta)^{10\delta_{\min}} a^{15\delta_{\min}+e_a}.$$

It remains to find (e_T, e_a). The definitions of S and H with $HAT \le 56a$ give $ST^3 \ll a$, and $HT^3 \le 448\Delta a$ follows from the first and third window inequalities. Since $H \ge 1$ and $m \ge 1$, $\max(H, S^m) \le \max(H, S)^m$. So if $\mathcal{B} = \max(H, S^m)$ we may take $(e_T, e_a) = (3m, m)$, up to divisor factors. The definition of Y gives $YT^{20} \ll \Delta^5 a^6$, so if $\mathcal{B} = Y^n$ we may take $(e_T, e_a) = (20n, 6n)$.

Both fractions in $E(m,n)$ exceed $5/17$, and $T^4 \ll a$ is stronger still. After fixing m, n, all divisor and logarithmic losses are $O(a^\eta)$ for every $\eta > 0$. So every window has $T \ll a^{E(m,n)+\varepsilon/2}$, and summing over the $O(\log a)$ windows proves the bound. For $m = n = 1$ the two fractions are $256/870 = 128/435$ and $261/887$, and the first is larger. For $(m,n) = (50, 51)$ the larger fraction is $2201/7480$. □

Proof of Theorem 1.1. Set $r = n/m$. The two fractions in $E(m,n)$ become

$$E_S(r) = \frac{150r + 106}{510r + 360}, \qquad E_Y(r) = \frac{156r + 105}{530r + 357}.$$

The first decreases and the second increases on $r > 0$. Their crossing satisfies $30r^2 - 10r - 21 = 0$, so $r_* = (5+\sqrt{655})/30$ and their common value is $\alpha_* = (1310 - \sqrt{655})/4365$. Positive rational ratios approximate r_*; choose the fixed positive integers m, n to make $E(m,n) < \alpha_* + \varepsilon/2$, and apply Corollary 15.16 with loss $\varepsilon/2$. The powers and all constants depend only on ε, not on a, b. □

Remark 15.17. In terms of the scales, at the critical point $T \approx a^{\alpha_*}$ one has $A \approx a^{2\alpha_*}$, $H \approx S \approx a^{1-3\alpha_*} \approx a^{0.117}$, $X \approx a^{3-10\alpha_*} \approx a^{0.057}$ and $Y \approx a^{6-20\alpha_*} \approx a^{0.115}$. The pattern count $X^5 \approx a^{0.287}$ alone would give the exponent $5/17$. The graph term multiplies it by $\mathcal{B}^{1/\delta_{\min}}$, and at $n/m \approx r_*$ the two box radii S^m and Y^n balance, at $\mathcal{B} \approx a^{0.117m}$. This gives the extra factor $a^{0.117/(10r_*+7)} \approx a^{0.0068}$.

16. The Lean formalisation

The accompanying Lean 4 project uses toolchain `v4.35.0` and Mathlib `v4.35.0`. It proves the main results of this paper, including the curve-counting input to Theorem 1.1. The logarithmic refinement is the one exception: its Lean proof takes the two published analytic estimates (9) and (10) as explicit hypotheses.

The strongest bound has the following statement:

```
theorem P_le_joint_optimal :
    ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ,
    0 < b → b < a → (P a b : ℝ) ≤
    C * (a : ℝ) ^ (((1310 : ℝ) - Real.sqrt 655) / 4365 + ε)
```

The order of the quantifiers makes the uniformity explicit: C may depend on ε, but not on a or b. Here `P a b` is defined by well-founded recursion,

$$P(a,0) = 0, \qquad P(a,b) = 1 + P(a, a \bmod b) \quad (b > 0),$$

so it counts exactly the positive terms of the orbit. The formal orbit is indexed from zero, as in this paper.

The following table gives the public declarations; namespaces and the `.lean` extension are omitted. The repository README gives the fuller map from proof steps to modules.

Result	Declaration	Module
Theorem 1.1, α_*	`P_le_joint_optimal`	`JointLiftMain`
Theorem 1.1, 2201/7480	`P_le_2201_7480`	`JointLiftMain`
Theorem 1.1, 128/435	`P_le_128_435`	`JointLiftMain`
Theorem 1.2	`P_le`	`Main`
Theorem 1.5	`P_le_23_77`	`TwentyThreeMain`
Theorem 1.6	`P_le_11_37`	`ElevenMain`
Theorem 1.7	`P_le_8_27`	`EightMain`
Theorem 1.8	`P_le_45_152`	`FortyFiveMain`
Theorem 1.3	`P_le_loglog`	`Hooley`
Theorem 1.4	`P_le_rises`	`RisesMain`
Theorem 10.2	`hankelInv_pos_int`	`Hankel`

The identifications of the first four invariants in Theorem 10.2 are proved in the same module as `hankelInv_one` through `hankelInv_four`.

For the curve theorem (28), the separate library `ccdn-dimension-growth` proves the real-B version, including the plane estimate, degree-preserving projection and box rescaling. The Pierce development imports it as a proved theorem and applies it to the defining prime ideals of its curves, using the same affine Hilbert degree as in Section 15. Thus the curve count is a proved dependency, not a hypothesis of the Pierce bound.

The proof of Theorem 1.1 is assembled in `PoweredJointBound`. The identities of Lemma 15.1(1) are proved in `FixedDividendIdentity` and `FixedDividendCurveEquations`, and the cover by at most twenty-four curves in `FixedDividendComponentCount`. The other main ingredients are the parity cover, the ordinary and conic gap counts, the graph-degree lower bounds, the powered pattern count and the window estimates; the repository README maps them to their modules. The transitive proof dependencies of the three public statements in `JointLiftMain` are audited to exclude the weighted-lattice counting modules of Section 14. In contrast, `P_le_loglog` takes the propositions `NairTenenbaumDelta` and `BretecheTenenbaumMean`, which state (9) (with $\sigma = 1/4$) and (10), including their existential constants and thresholds.

The file `Axioms.lean` audits the eleven public statements and their proof dependencies. The only logical axioms reported are `propext`, `Classical.choice` and `Quot.sound`, the standard axioms used by Mathlib. The proof library contains no `sorry` and uses no `native_decide`. The separate Mathlib-only `Challenge.lean` states the eleven results with deliberate theorem holes for comparison with the proved `Solution.lean`; neither module imports the other. Build instructions and the scope of the verification checks are recorded in the README.

Many local identities are checked by polynomial normalization over $\mathbb{Z}$, then transferred to the natural-number orbit after proving that the relevant differences are nonnegative. The counting arguments bound the number of parameter values and then the number of states with each value. Determinant positivity, integrality, curve reconstruction and the valuation arguments are proved within the same development. The divisor estimate in Lemma 2.2 uses `DivisorPowerBound.lean` from `openai/math` [10] (Apache-2.0), with unchanged proof bodies and adapted imports and module visibility. The required Hadamard inequality is proved in `Hadamard.lean`.

17. Limits of the method

The threshold 5/17. In the proof of Corollary 15.16, three contributions give exactly the exponent 5/17, up to the arbitrarily small loss: the pattern count X^5 (from components that

are not retained and from conics), the exceptional patterns, and windows in which the line-exclusion condition (21) fails. None of them involves the added coordinates W and J. So improving the coordinate embedding alone cannot go below $5/17 = 0.2941176\ldots$, which α_* exceeds by about $1.3 \cdot 10^{-4}$. A stronger degree bound for the powered graphs may bring the exponent closer to $5/17$; going below it requires a sharper count of the patterns that actually occur along an orbit. No claim of optimality for the Pierce iteration is made.

The earlier arguments. Proposition 7.3 removes the divisor loss from the local bound, but on its own the exponent is still $3/10$. The local bound $T \ll a^{1/2}A^{-1/3}$ and the drop bound $T \ll A^{1/2}$ (up to the divisor factor) both allow $T \asymp a^{3/10}$ at $A \asymp a^{3/5}$. Even an ideal short-interval divisor estimate with G constant leaves this balance unchanged. So sharpening constants or improving the divisor counts in that argument cannot give a logarithmic bound. To improve that argument, one needs to restrict *which* configurations of the parameters (u, v, w, z, D, E, F) can occur together along a single orbit. Theorem 1.4 is one restriction of this kind: a long orbit must have many rises in its quotient increments. Theorem 10.2 supplies the natural candidates for further ones, the higher invariants I_r. Section 11 uses I_4 in exactly this way.

The count of Section 12 loses through the number of patterns, X^7, reduced to X^5 in Section 13 and weighted by the lattices Λ_p in Section 14. With patterns at four consecutive states one has thirteen integers and as many equations as unknown increments, so one might hope for $O(1)$ configurations per pattern. But the four-state equations have positive-dimensional families of solutions. For instance, the Newton coefficients $(\alpha, -3\alpha-1, 3\alpha+3, -\alpha-3, 1)$, with coefficient polynomial $(z-1)^3(z-\alpha)$, at every state, together with constant increments $k_j = t$, satisfy all the linear and quadratic relations for every t. Such families are excluded only by the remainder equations, which those earlier counts discard. The fixed-dividend curve equations of Section 15 retain this information. Section 14 uses only the first minimum of Λ_p, and slices in a single direction; the other two minima and the shape of the quadric in the lattice are not used.

There is no obvious monotonicity to exploit. The curvatures D_i of successive blocks need not decrease. For $a = 13242$ the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132,\ \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$. The curvatures of the first five blocks are $24, 12, 6, 111, 30$, all for blocks that lie within a factor of two.

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